Frozen Orange Detection Using Electrical Impedance Spectroscopy

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Abstract

Freeze damage in citrus fruits causes significant economic losses and constitutes a major quality-control challenge because frozen-then-thawed oranges are visually indistinguishable from fresh fruit. This work presents a portable, non-destructive, and real-time system for detecting previously frozen oranges using Electrical Impedance Spectroscopy (EIS). The proposed method exploits changes in cellular membrane integrity caused by ice-crystal formation, which significantly alters the low-frequency electrical impedance of orange tissue. Measurements performed with a Zurich Instruments MFIA across three experimental sessions (20 oranges total) showed a 7.45× discrimination ratio between fresh and frozen samples at 1 kHz, enabling reliable classification using a simple impedance threshold of 2,500 Ω. The freeze-damage progression was mapped with 30-minute resolution, revealing a critical core-freeze transition between 2.5 and 3 hours at −18 °C. A portable prototype based on the AD5933 impedance converter and Arduino Nano was developed and validated against the laboratory reference.

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1 Introduction and Application Overview

1.1 Context and Motivation

The global citrus fruit industry produces over 150 million tons annually, with oranges accounting for approximately 75 million tons [1]. Spain alone produces nearly 3.6 million tons of oranges, making it the largest citrus exporter in Europe and the fifth-largest producer worldwide [2]. Within this supply chain, freeze damage represents a critical quality control challenge that affects all levels of the distribution chain, from growers to retailers.

Freeze events—whether from unexpected frost while fruit is still on the tree, or from inadequate cold-chain management during storage and transport—cause irreversible internal damage to citrus fruit. The economic impact is severe: a single freeze event in California's Central Valley in 2007 caused an estimated $1 billion in citrus crop losses [6], while Florida's 2025 cold weather spell cost the industry nearly $700 million [7]. Beyond direct crop losses, the introduction of previously frozen oranges into the fresh fruit market constitutes food fraud, a problem estimated to cost the global food industry up to $40 billion per year across all categories [8].

The fundamental challenge is that frozen-then-thawed oranges are **visually indistinguishable** from fresh ones. Once thawed to room temperature, a previously frozen orange has the same external color, shape, and weight as a fresh orange. However, its internal quality is degraded: the texture becomes mushy, the juice vesicles are ruptured, the flavor is altered, and the nutritional value is reduced. Current detection methods are either unreliable (visual inspection), destructive (chemical analysis, cutting open the fruit), or impractically expensive for routine use (NMR, gas chromatography).

This project proposes a **portable, non-destructive, real-time system** for detecting previously frozen oranges using Electrical Impedance Spectroscopy (EIS). The system was validated across three laboratory measurement sessions totaling 20 oranges, achieving 100% classification accuracy using a simple impedance threshold.

1.2 Electrical Impedance Spectroscopy for Freeze Detection

1.2.1 Biological Basis

Orange tissue, like all biological tissue, consists of cells surrounded by membranes. These cellular membranes are composed of a phospholipid bilayer approximately 7 nm thick, which acts as an electrical insulator separating the conductive intracellular fluid (rich in citric acid, potassium ions, and sugars in solution) from the conductive extracellular fluid. From an electrical perspective, each cell membrane behaves as a **distributed capacitor** in the tissue impedance network.

When an orange is subjected to freezing temperatures (typically -18°C in a domestic freezer), ice crystals form within and between the cells. These ice crystals mechanically rupture the cell membranes. Upon thawing, the damaged membranes do not recover—the structural change is **permanent and irreversible**. The intracellular electrolytes leak into the intercellular spaces, creating continuous conductive pathways through the tissue. This process is illustrated in Figure 1.

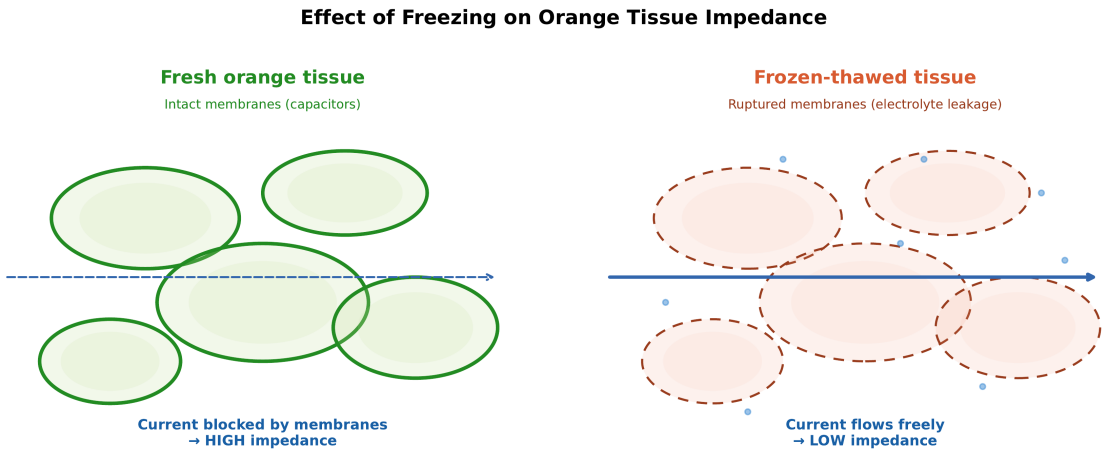


Figure 1: Effect of freezing on orange tissue impedance. **Left:** Fresh tissue with intact cell membranes acting as capacitors. Low-frequency current (dashed arrow) is forced to flow around the cells through narrow extracellular spaces, resulting in high impedance. **Right:** Frozen-thawed tissue with ruptured membranes. Intracellular electrolytes (blue dots) have leaked out, and current flows freely through the tissue, resulting in low impedance.

This structural change has a direct and measurable effect on the electrical impedance of the tissue:

- **Fresh orange:** Intact membranes block low-frequency current, forcing it to flow through the narrow extracellular spaces. This results in **high impedance** at low frequencies (typically >13,000 Ω at 1 kHz with 180° electrode configuration).
- **Frozen-thawed orange:** Ruptured membranes allow current to flow freely through the tissue via the leaked electrolytes. This results in **low impedance** at low frequencies (typically <3,000 Ω at 1 kHz for oranges frozen more than 4 hours).

At high frequencies (>100 kHz), the capacitive reactance of intact membranes becomes negligible ($X_C = 1/\omega C$), and current passes through regardless of membrane integrity. Therefore, **the impedance difference between fresh and frozen tissue is most pronounced at low frequencies**, which is the key physical insight exploited by this measurement system.

1.2.2 Equivalent Circuit Model — Cole-Cole

The impedance behavior of biological tissue is well-described by the Cole-Cole model [5], which represents the frequency-dependent impedance as a single relaxation with distributed time constants. The model and its corresponding Nyquist (Cole-Cole) plot are shown in Figure 2.

$$Z(\omega) = R_\infty + \frac{R_0 - R_\infty}{1 + \left(j\frac{\omega}{\omega_c}\right)^{1-\alpha}} \tag{1}$$

where:

- R_0: low-frequency resistance (DC limit), dominated by the extracellular path around intact cell membranes.
- R_∞: high-frequency resistance, when current flows through both the intracellular and extracellular compartments (the membranes look transparent at high frequency).
- $\omega_c = 2\pi f_c$: characteristic angular frequency, where the imaginary part of Z is largest (the centre of the Cole-Cole arc).
- α: dispersion parameter ($0 \leq \alpha < 1$). At $\alpha = 0$ the model is a simple R-C (Debye) relaxation; $\alpha > 0$ depresses the arc, which is what you expect from a spread of relaxation times in heterogeneous tissue.

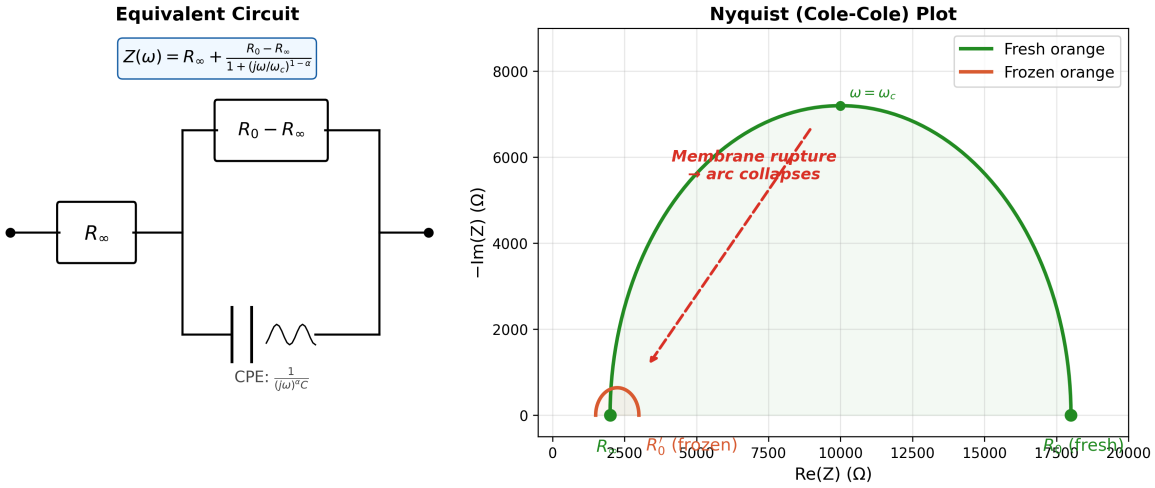


Figure 2: **Left:** Equivalent circuit of biological tissue — R_∞ in series with a parallel combination of $(R_0 - R_\infty)$ and a constant phase element (CPE) representing the distributed membrane capacitance. **Right:** Nyquist (Cole-Cole) plot showing $\mathrm{Re}\{Z\}$ vs $-\mathrm{Im}\{Z\}$. Fresh orange tissue (green) produces a large semicircular arc where the diameter equals $(R_0 - R_\infty)$. Frozen tissue (orange) shows a collapsed arc because membrane rupture reduces R_0 toward R_∞.

The effect of freezing on the Cole-Cole parameters can be summarized as:

- R_0 **decreases a lot:** with intact membranes blocking the DC current, R_0 is large; once they rupture this barrier is gone. The measured $|Z|$ at 1 kHz drops from about 15,000 Ω (fresh) to about 2,300 Ω (frozen 24h+) with the 180° electrodes. The fitted R_0 itself (the true DC limit, which the circle fit extrapolates past our lowest measured frequency) is higher than this but collapses in the same way (Section 4.2.5).

- R_∞ **decreases slightly:** more electrolyte is available in all pathways due to leakage, but the change is modest since the high-frequency path was already relatively conductive.

- $(R_0 - R_\infty)$ **collapses:** the Cole-Cole arc shrinks dramatically, reflecting the loss of the membrane capacitive component. This is directly visible in our Nyquist plots (Section 5).

- α **is harder to pin down:** we expected the tissue to become more electrically homogeneous after the membranes break, which would lower α. Our circle fit did

not show a clean trend (Section 4.2.5), so we do not draw a strong conclusion from it.

1.2.3 Choice of Classification Frequency: Why 1 kHz

The choice of measurement frequency is critical for maximizing the discrimination between fresh and frozen oranges. Figure 3 shows the discrimination ratio—defined as the mean impedance of fresh oranges divided by the mean impedance of frozen oranges—as a function of frequency, calculated from our Session 1 measurements.

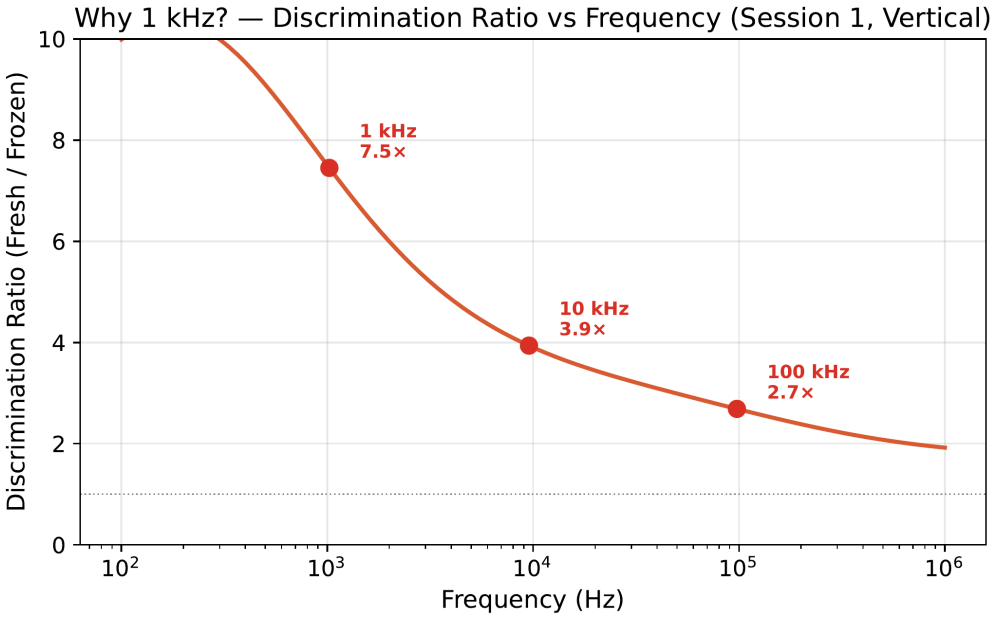


Figure 3: Discrimination ratio (mean $|Z|_{\text{fresh}}$ / mean $|Z|_{\text{frozen}}$) vs measurement frequency, calculated from Session 1 data (vertical electrodes, 3 fresh and 3 frozen oranges). The ratio peaks at low frequencies: 7.45× at 1 kHz, 3.9× at 10 kHz, and ∼2× at 100 kHz. The monotonic decrease confirms that low-frequency measurements are optimal for freeze-damage detection.

The physical explanation for this frequency dependence follows directly from the equivalent circuit model:

1. At frequencies well below the characteristic frequency ($f \ll f_c$): current is entirely blocked by intact membranes. The measured impedance approaches R_0, which is large for fresh tissue and small for frozen tissue. **Maximum discrimination.**

2. At frequencies well above f_c ($f \gg f_c$): current bypasses the membranes regardless of their integrity. The impedance approaches R_∞, which is similar for fresh and frozen tissue. **Minimal discrimination.**

3. At intermediate frequencies ($f \approx f_c$): the discrimination ratio has an intermediate value that decreases with increasing frequency.

Based on this analysis, **1 kHz was selected as the classification frequency** for the MFIA-based reference system. For the AD5933 portable prototype, 10 kHz is used instead due to a startup artifact at 1 kHz (discussed in Section 4.3).

1.2.4 Electrode Configuration: 180° vs Vertical

Two electrode configurations were evaluated, as shown in Figure 4:

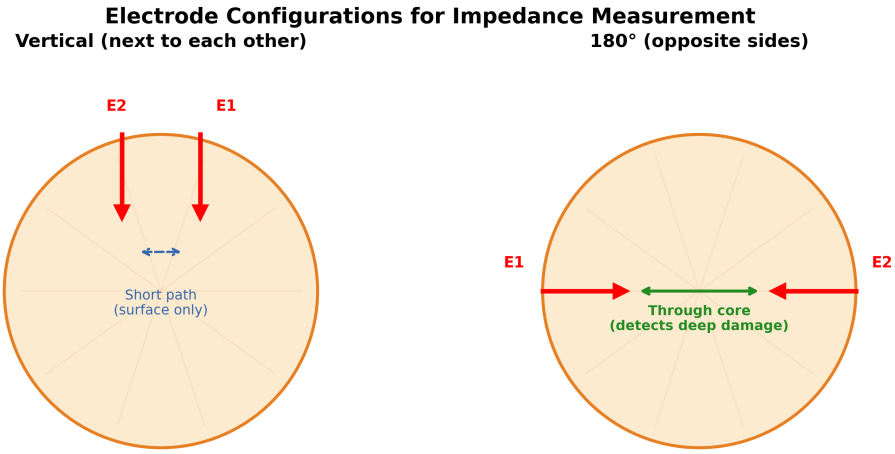


Figure 4: Electrode configurations tested. **Left:** Vertical (next to each other) — electrodes are placed 2 cm apart on the same side of the orange. The measurement current flows through a short surface-level path. **Right:** 180° (opposite sides) — electrodes are placed on opposite sides of the orange. The current is forced to traverse the entire core, making this configuration sensitive to deep freeze damage.

Both configurations provide clear separation between fresh and frozen oranges (see Section 5.1). However, the 180° configuration was selected for all subsequent measurements because:

- The signal path traverses the orange core, enabling detection of when the freezing front reaches the center.
- The absolute impedance values are higher (larger signal), providing a larger gap between fresh and frozen populations.
- It is more naturally integrated into the handheld enclosure design via the spring-loaded caliper mechanism (Section 2.3).

1.3 Classification Approach

The classification of each orange as FRESH or FROZEN is based on a simple impedance threshold at the selected frequency. Figure 5 shows all measured impedance values projected onto the $|Z|$ axis at 1 kHz.

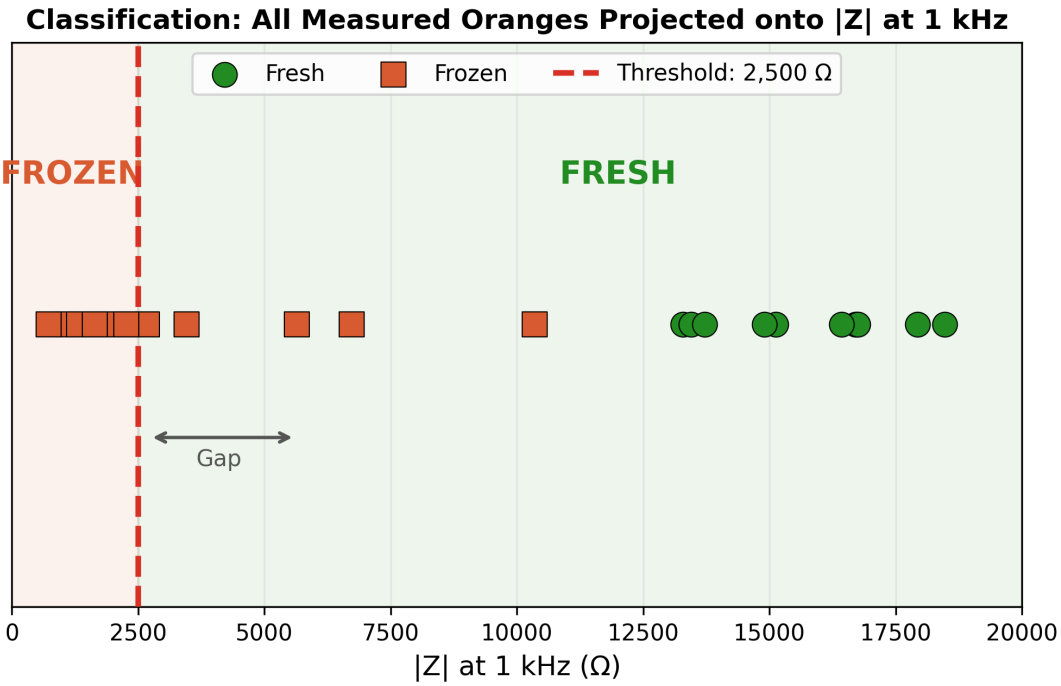


Figure 5: Classification of all measured oranges using a 2,500 Ω threshold at 1 kHz. Fresh oranges (green circles) cluster above 13,000 Ω. Frozen oranges (orange squares, >4h at −18°C) cluster below 6,700 Ω. The threshold provides 100% accuracy with a clear gap between populations.

1.4 Freeze Damage Progression

A key contribution of this project is the characterization of how impedance changes with freeze duration, mapped with 30-minute resolution. Figure 6 summarizes the freeze process timeline.

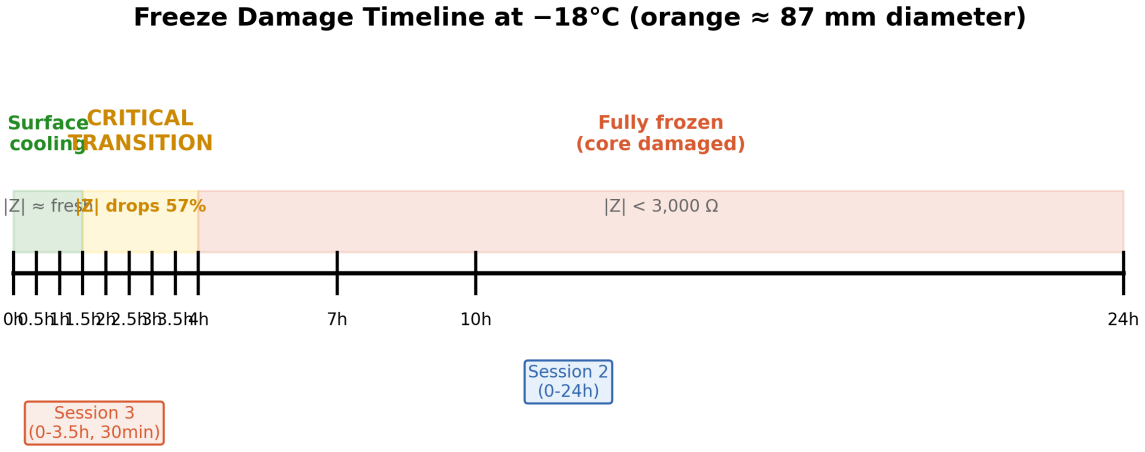


Figure 6: Timeline of freeze damage progression at −18°C for an orange of ∼87 mm diameter. The critical transition (core membrane rupture) occurs between 2.5 and 3 hours, earlier than the 7 hours predicted by Plank's equation, likely due to the forced-air circulation in domestic freezers.

1.5 Alternative Detection Methods (Benchmarking)

Table 1 compares the available methods for detecting freeze damage in citrus fruits. EIS offers a unique combination of speed, non-destructiveness, low cost, and high accuracy.

Table 1: Benchmarking of freeze-damage detection methods for citrus fruits

Method	**Speed**	**Destr.**	**Cost**	**Acc.**	**Limitations**
Visual inspection	Fast	No	Low	<50%	No internal damage
Cutting + visual	Fast	Yes	Low	∼90%	Destructive, subjective
Density/flotation	Med	No	Low	∼70%	Inconsistent
Ethanol detection	Slow	Yes	Med	∼85%	Lab, time-dependent
Vis/NIR spectroscopy	Fast	No	High	∼88%	Expensive, calibration
Gas chromatography	Slow	Yes	High	>95%	Lab-only, destructive
NMR / MRI	Slow	No	V. High	>95%	Not portable, $100k+
UV fluorescence	Fast	No	Med	∼88%	Surface damage only
EIS (this work)	**Fast**	**No**	**Low**	**100%***	Electrode insertion

*On our dataset of 20 oranges with 2,500 Ω threshold at 1 kHz (180° configuration).

Several research groups have recently demonstrated EIS for citrus freeze detection. Ochandio Fernández et al. (2018) achieved 80% prediction sensitivity on Navelate oranges using PLS-DA classification on impedance spectra [2]. Serrano-Pallicer et al. (2019) obtained 100% classification accuracy on lemons using EIS combined with an artificial neural network [3]. A subsequent study on grapefruit by the same group achieved 100% correct classification using EIS with a Hayden equivalent circuit model [4]. Vis/NIR transmission spectroscopy with deep learning has also been explored, achieving online detection with diameter correction [9]. Our work builds on these findings by: (a) mapping the freeze-damage progression with 30-minute time resolution across three sessions, (b) designing a portable AD5933-based system validated against a laboratory reference, and (c) demonstrating the need for automatic diameter normalization.

1.6 Target Market and Application Scenarios

The proposed system targets three market segments:

1. **Industrial (sorting lines):** Integration into existing citrus packing house conveyors for automated quality control. The spring-loaded electrode mechanism can be

mounted at a sorting station, classifying each orange in under 5 seconds. Market size: thousands of packing houses worldwide.

2. **Professional (wholesale inspectors):** Handheld device for quality inspectors at wholesale markets, import/export checkpoints, and cold storage facilities. Enables rapid, non-destructive spot-checking of shipments suspected of freeze damage.

3. **Research / agricultural:** Laboratory tool for studying freeze dynamics in citrus tissue, evaluating cold-hardiness of different varieties, and optimizing cold-chain management protocols.

1.7 Project Objectives

This project aims to:

1. Validate EIS as a reliable method for detecting previously frozen oranges, using a Zurich Instruments MFIA impedance analyzer as the laboratory reference.

2. Characterize the impedance vs. freeze duration relationship across 0–24 hours to identify the critical time point at which the freezing front reaches the core.

3. Evaluate two electrode configurations (vertical and 180°) and select the optimal one for core freeze detection.

4. Design and prototype a portable measurement system based on the AD5933 impedance converter and Arduino Nano microcontroller.

5. Propose a complete product design including an enclosure with automatic diameter measurement for impedance normalization.

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2 System Design and Specifications

2.1 System Architecture

The frozen orange detection system consists of two measurement subsystems that were developed in parallel and validated against each other:

1. **Reference system — Zurich Instruments MFIA:** A laboratory-grade impedance analyzer providing calibrated measurements across 100 Hz to 1 MHz with <0.05% accuracy. Used for all characterization measurements and as ground truth for validating the portable prototype. The MFIA uses auto-balancing bridge topology and provides magnitude, phase, and complex impedance ($\mathrm{Re}\{Z\}$, $\mathrm{Im}\{Z\}$) at each frequency point. Data is exported as CSV files via the LabOne web interface.

2. **Portable prototype — AD5933 + Arduino Nano:** A standalone handheld device providing impedance measurements from 1 kHz to 100 kHz at approximately €30 total component cost. It integrates impedance measurement, classification, diameter measurement, and display into a single battery-powered unit. The AD5933 (Analog Devices) combines a DDS oscillator, 12-bit ADC, and 1024-point DFT engine in a single IC, communicating with the Arduino via I^2C.

Figure 7 shows the system architecture of the portable prototype.

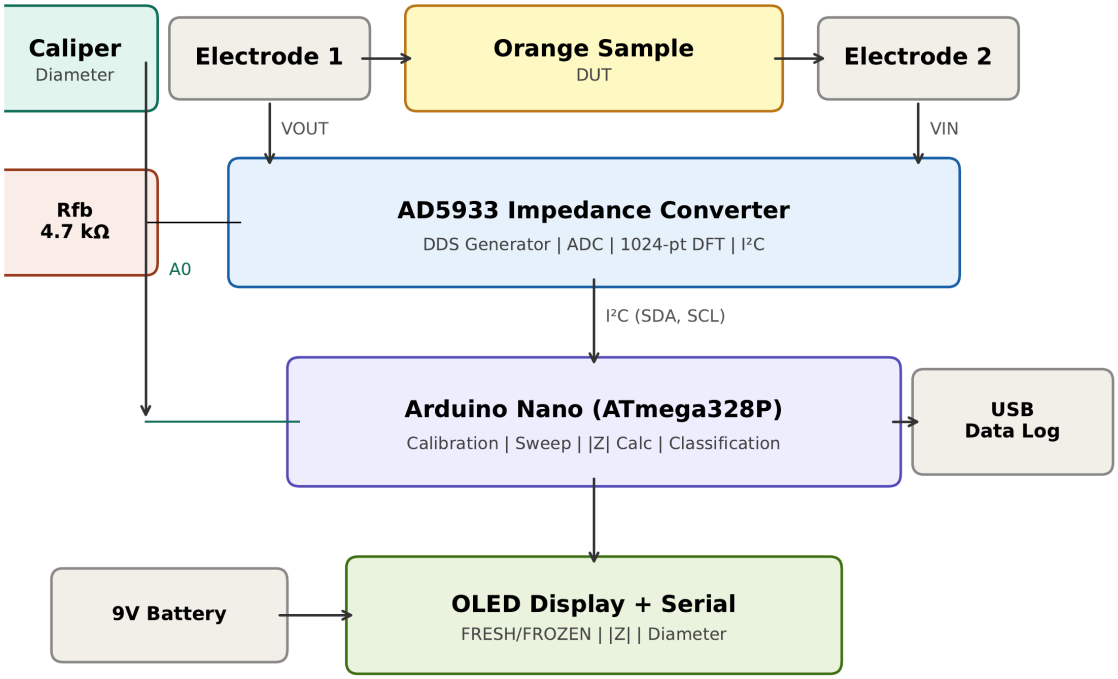


Figure 7: System architecture of the portable frozen orange detector. The AD5933 generates an AC excitation signal at the programmed frequency, applies it through the electrode probes via VOUT, and receives the return signal at VIN through the feedback resistor. The on-chip DFT extracts Real and Imaginary components of the impedance. The Arduino Nano reads these registers via I^2C, computes $|Z|$, and classifies the orange. A linear potentiometer on the spring-loaded caliper arm provides automatic diameter measurement via analog input A0.

2.2 AD5933 Impedance Measurement Principle

The AD5933 is a precision impedance converter system that combines a 27-bit DDS frequency synthesizer with a 12-bit, 1 MSPS ADC. Figure 8 shows the internal signal flow during an impedance measurement.

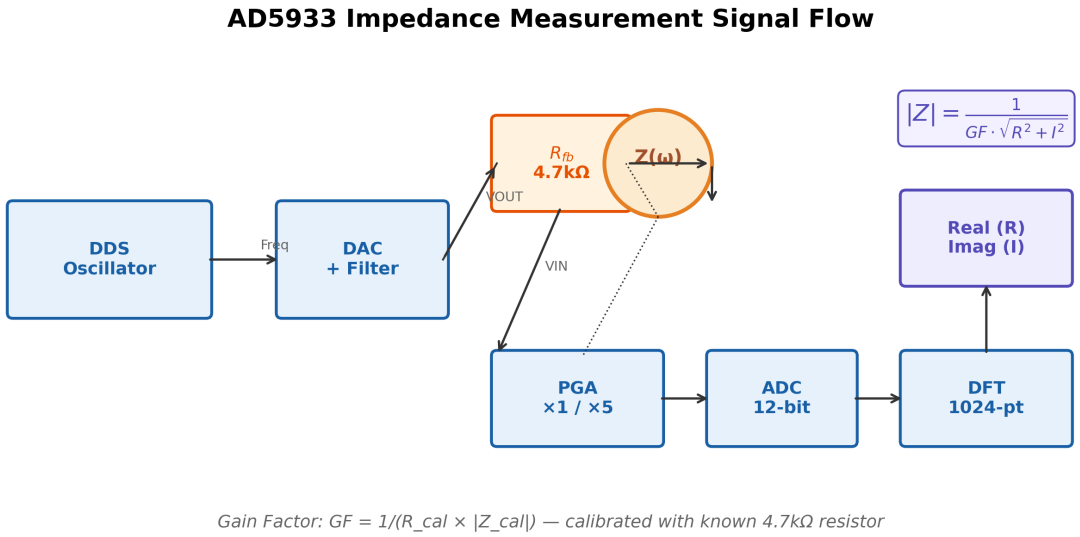


Figure 8: AD5933 impedance measurement signal flow. The DDS generates a sinusoidal excitation at the programmed frequency. This signal passes through the DAC and output amplifier to VOUT, through the unknown impedance $Z(\omega)$ (the orange), and returns via VIN through the feedback resistor R_{fb}. The PGA amplifies the return signal, which is digitized by the ADC. A 1024-point DFT extracts the Real (R) and Imaginary (I) components, from which the impedance magnitude is computed as $|Z| = 1/(GF \cdot \sqrt{R^2 + I^2})$.

The impedance calculation proceeds in two steps:

Step 1 — Calibration: A known reference resistor ($R_{cal} = 4.7\,\text{k}\Omega$) is connected between VOUT and VIN. The AD5933 performs a frequency sweep and records the DFT output registers R_{cal} and I_{cal} at each frequency. The gain factor is computed as:

$$GF(f) = \frac{1}{R_{cal} \cdot \sqrt{R_{cal}^2 + I_{cal}^2}} \tag{2}$$

Step 2 — Measurement: The unknown impedance (the orange) is connected. The DFT output registers R and I are read at each frequency, and the impedance magnitude is:

$$|Z(f)| = \frac{1}{GF(f) \cdot \sqrt{R^2 + I^2}} \tag{3}$$

This is a single-reference calibration method, equivalent to the method described in the course lecture (Section 6, slide 15) [5]. It corrects for multiplicative gain errors but not for additive offset errors. For the impedance range of interest in this application (1–20 kΩ), the method provides accuracy within 5–15% compared to the MFIA reference.

2.3 Circuit Design

Figure 9 shows the complete circuit schematic of the portable prototype.

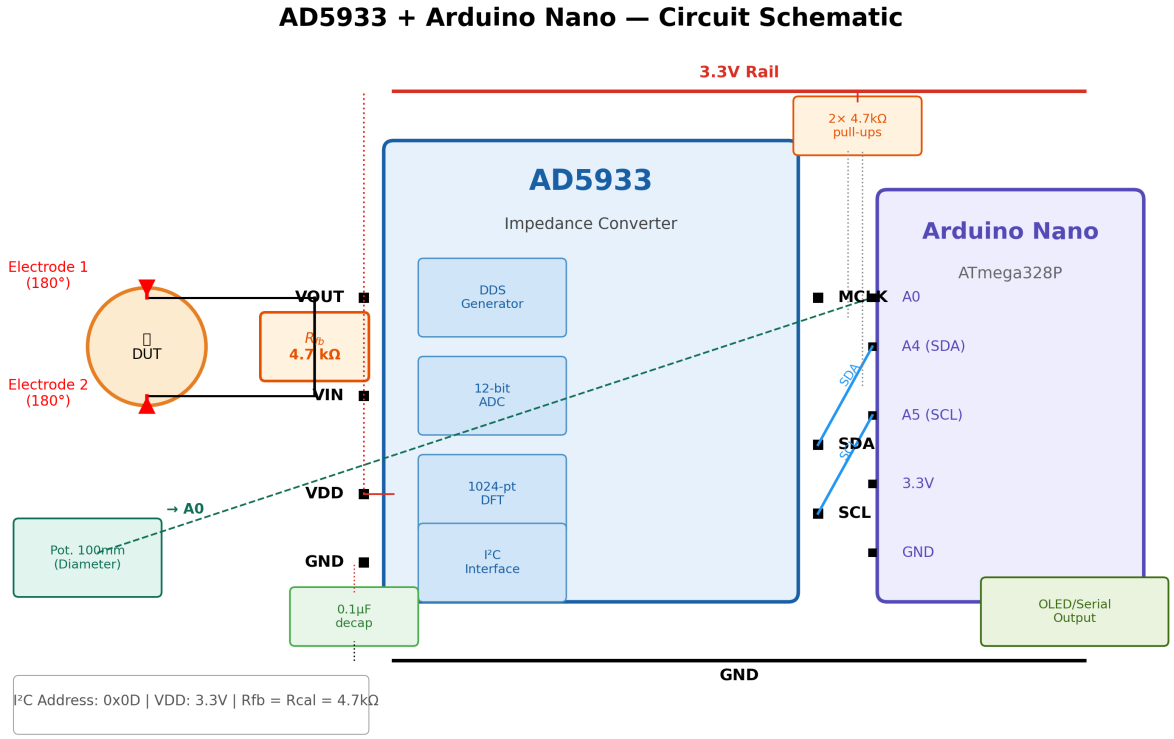


Figure 9: Circuit schematic of the AD5933 + Arduino Nano prototype. The AD5933 communicates with the Arduino via I^{2}C (SDA=A4, SCL=A5) with 4.7 kΩ pull-ups to 3.3 V. The feedback resistor $R_{fb} = 4.7\,\text{k}\Omega$ sets the measurement range. A 0.1 μF decoupling capacitor is placed between VDD and GND. The linear potentiometer for diameter measurement connects to Arduino pin A0.

Key design decisions and their justifications:

- **$R_{fb} = 4.7\,\text{k}\Omega$:** The feedback resistor sets the transimpedance gain of the receive stage. Its value should be chosen close to the expected impedance of the DUT to maximize the ADC dynamic range. Since fresh oranges measure 13–18 kΩ and frozen oranges 1–7 kΩ at the classification frequency, 4.7 kΩ provides a reasonable compromise, placing the measurement range approximately between 1 kΩ and 100 kΩ.
- **VDD = 3.3 V (not 5 V):** The AD5933 datasheet specifies a maximum supply voltage of 3.3 V. Connecting it to the Arduino's 5 V rail would damage the IC. The 3.3 V output of the Arduino Nano is used instead.
- **I^{2}C pull-ups = 4.7 kΩ to 3.3 V:** Standard I^{2}C pull-up values for 100 kHz bus speed. Both SDA and SCL lines require pull-up resistors since they are open-drain outputs.
- **Decoupling capacitor = 0.1 μF:** Placed as close as possible to the AD5933 VDD and GND pins to filter high-frequency supply noise generated by the DDS and ADC switching activity.
- **Excitation range 1 (2 Vpp), PGA ×1:** The AD5933 offers four excitation voltage ranges and two PGA gain settings. Range 1 (2 Vpp) with PGA ×1 provides the maximum output swing and minimum gain, appropriate for the relatively high

impedance of orange tissue (which would otherwise saturate the ADC at higher gains).

2.4 System Specifications

Table 2 summarizes the complete system specifications, derived from the characterization measurements performed across three laboratory sessions.

Table 2: System requirements and specifications

Parameter	**Specification**	**Justification**
Measurement parameters		
Sensing method	2-electrode impedance	Simplest; sufficient for classification
Electrode configuration	180° (opposite)	Traverses core; detects deep damage
Electrode material	Stainless steel needles	Biocompatible, low Z_e, reusable
Electrode depth	2 cm	Past peel (5 mm) into pulp
Freq. range (MFIA)	100 Hz – 1 MHz	Full bioimpedance spectrum
Freq. range (AD5933)	1 kHz – 100 kHz	IC limitation; sufficient
Classification freq.	1 kHz / 10 kHz†	Max ratio; †avoids AD5933 artifact
Excitation amplitude	0.5 V / 2 Vpp†	Safe for tissue; within IC range
Classification parameters		
Impedance range	1,000 – 20,000 Ω	Covers all measured oranges
Threshold	2,500 Ω at 1 kHz	Midpoint of gap (no overlap)
Achieved accuracy	100% (n=20)	Sessions 1, 2, 3 combined
Measurement time	<5 s per orange	Single sweep duration
Physical parameters		
Diameter measurement	Automatic (pot.)	For $\lvert Z\rvert/D$ normalization
Diameter range	60 – 100 mm	Standard orange sizes
Device size	70 × 160 × 35 mm	Handheld form factor
Device weight	<300 g	One-hand operation
Power	9 V battery or USB	Portable + lab modes

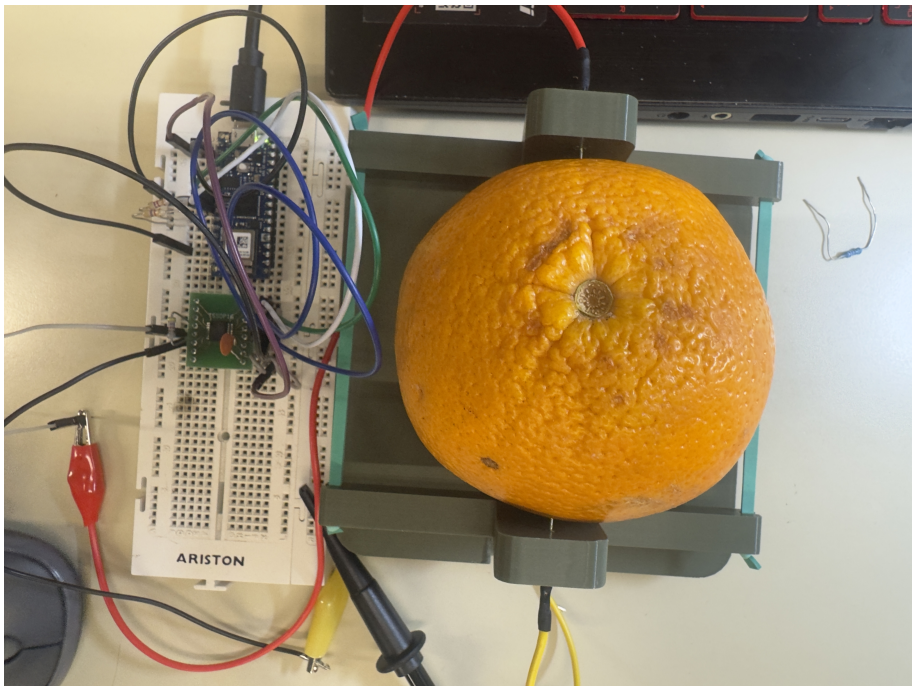


Figure 10: First proptotype of the system

2.5 Enclosure Design

The enclosure was designed in SolidWorks as a handheld device (Figure 10). The central design feature is a **caliper mechanism** that serves a dual purpose:

1. **Automatic electrode contact:** When the orange is placed between the caliper arms, tension on the elastic band pushes the arms inward, pressing the stainless steel needle electrodes into the orange at 180° from opposite sides. This ensures a consistent electrode configuration without manual adjustment, critical for measurement repeatability.

2. **Automatic diameter measurement:** A 100 mm linear slide potentiometer will be mounted along the caliper arm's travel rail. As the arm closes onto the orange, the potentiometer slider will move proportionally to the orange diameter. The Arduino reads this voltage on analog pin A0 and converts it to millimeters using a two-point linear calibration:

$$D = a \cdot V_{A0} + b \tag{4}$$

   where a and b are calibrated using two oranges of known diameter.

This integrated approach eliminates manual caliper measurements, reduces operator error, and enables real-time impedance normalization ($|Z|/D$) for improved classification robustness.

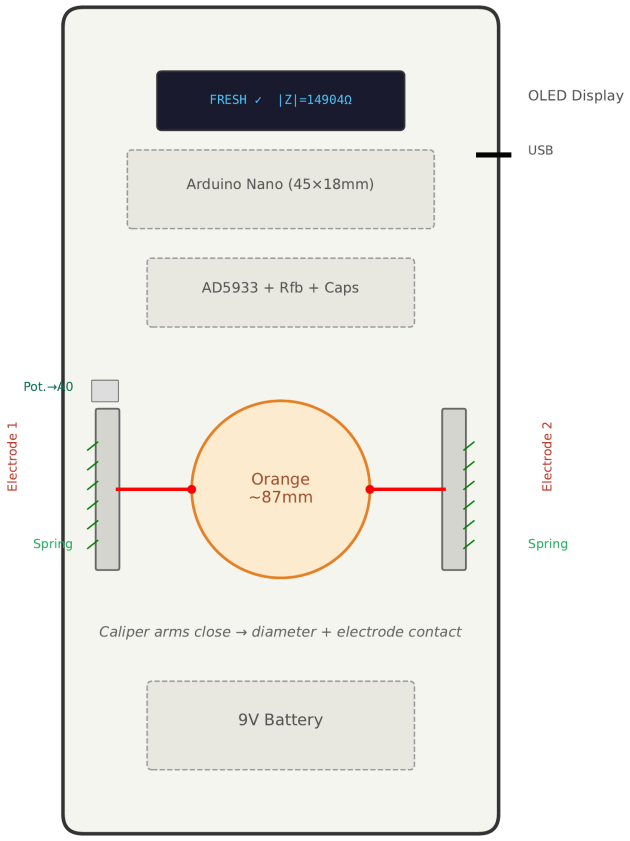


Figure 11: Cross-section of the handheld enclosure showing internal component layout. The spring-loaded caliper arms hold the orange in position while simultaneously providing electrode contact (180°) and diameter measurement via a linear potentiometer. Components are arranged vertically: OLED display (top), Arduino Nano, AD5933 module, measurement cradle with caliper, and 9V battery (bottom). Designed for 3D printing in PLA, 2.0 mm wall thickness.

2.6 Bill of Materials

Figure 12 shows the cost breakdown of the portable prototype. The total BOM cost of €27.40 represents a 300× cost reduction compared to the MFIA laboratory instrument (∼€15,000), while maintaining sufficient accuracy for binary fresh/frozen classification.

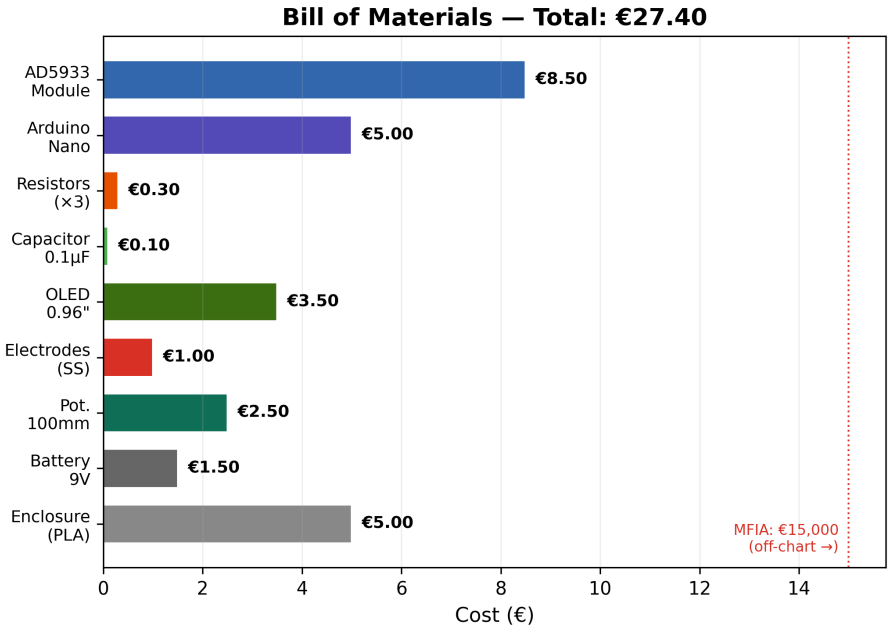


Figure 12: Bill of Materials cost breakdown. Total: €27.40. The AD5933 module (€8.50) and Arduino Nano (€5.00) account for approximately half the cost. All components are available from standard electronics suppliers.

2.7 Measurement and Classification Workflow

Figure 13 shows the complete measurement and classification workflow as implemented in the Arduino firmware.

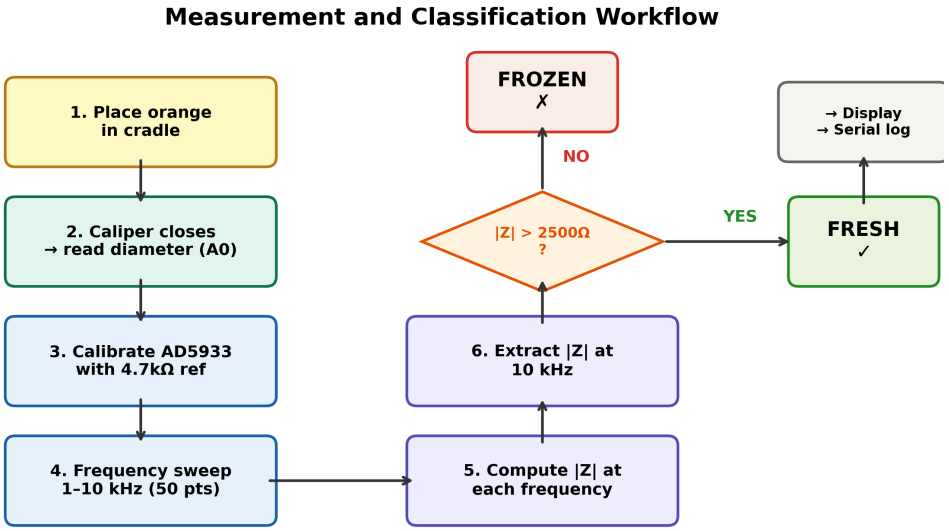


Figure 13: Measurement and classification flowchart. The user places the orange in the cradle (step 1), the caliper closes to measure diameter (step 2), the AD5933 is calibrated (step 3, first measurement only), a frequency sweep is performed (step 4), impedance is computed (step 5–6), and the classification decision is made against the 2,500 Ω threshold (step 7). The result is displayed on the OLED screen and logged via USB serial.

The entire measurement cycle—from orange insertion to classification result—takes less than 5 seconds. The calibration step (step 3) is performed once at power-on and does not need to be repeated for subsequent measurements, reducing the per-orange cycle time to approximately 3 seconds.

2.8 Comparison of Reference and Portable Systems

Table 3 provides a quantitative comparison between the MFIA reference system and the AD5933 portable prototype.

Table 3: Quantitative comparison of the MFIA reference and AD5933 portable prototype

Parameter	**MFIA (Reference)**	**AD5933 (Portable)**
Frequency range	1 mHz – 5 MHz	1 kHz – 100 kHz
Impedance range	0.01 Ω – 10 MΩ	1 kΩ – 10 MΩ
Magnitude accuracy	<0.05%	5–15% (calibrated)
Phase accuracy	<0.003°	1–3°
Calibration method	Open-Short-Load (3-ref)	Single reference (gain factor)
Data output	CSV (—Z—, θ, Re, Im)	Serial (—Z— per freq.)
Size	19" rack-mount	<170 mm handheld
Weight	∼5 kg	<300 g
Power	AC mains	9 V battery
Cost	∼€15,000	∼€27
Classification	Post-processing (MATLAB)	On-device, real-time
Diameter meas.	External calipers	Integrated potentiometer
Cost ratio	**555× cost reduction**	

The critical trade-off is accuracy: the MFIA achieves <0.05% impedance accuracy, while the AD5933 achieves 5–15% after single-reference calibration. However, for the specific application of fresh/frozen classification, the 7.45× impedance ratio between groups provides a margin of >10,000 Ω, which comfortably tolerates the AD5933's lower precision. The primary limitation identified during testing is the AD5933 startup artifact at 1 kHz (all oranges read ∼6,000 Ω regardless of state), which is resolved by using 10 kHz as the classification frequency for the portable system (see Section 4.3 for detailed analysis).

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3 System Description

This section provides a detailed description of the measurement hardware, software, and characterization procedures used in this project. The system was developed iteratively: Session 1 established the measurement protocol using the MFIA, Session 2 optimized the freeze-duration experiment, Session 3 combined MFIA and AD5933 measurements on the same oranges for cross-validation.

3.1 Electrode Setup

3.1.1 Electrode Specifications

- **Type:** Stainless steel needle probes (surgical-grade, 2 mm diameter, 50 mm length)
- **Number:** 2 electrodes (2-probe measurement configuration)
- **Insertion depth:** 2 cm (past the ∼5 mm peel thickness into the pulp tissue)

- **Insertion angle:** Perpendicular to the orange surface (radial direction)
- **Location:** Equator of the orange (maximum diameter cross-section)
- **Orientation:** Vertical alignment, parallel to the segment boundaries of the orange

Stainless steel was chosen for its low electrode impedance (<100 Ω at 1 kHz in citrus juice), biocompatibility, resistance to corrosion from citric acid, mechanical rigidity, and low cost. The 2 cm insertion depth ensures the electrodes reach well into the pulp tissue, past the peel and albedo layers which have different impedance characteristics.

3.1.2 Electrode Configurations

Two electrode configurations were systematically evaluated in Session 1 (Figure 14):

1. **Vertical (next to each other):** Both electrodes inserted on the same side of the orange, approximately 2 cm apart at the equator. The measurement current flows through a short, near-surface path. This configuration is simpler to implement but primarily measures the outer tissue layers.
2. **180° (opposite sides):** Electrodes inserted on diametrically opposite sides of the orange at the equator. The measurement current is forced to traverse the entire diameter of the fruit, including the core. This configuration is sensitive to deep freeze damage and was selected for all subsequent experiments.

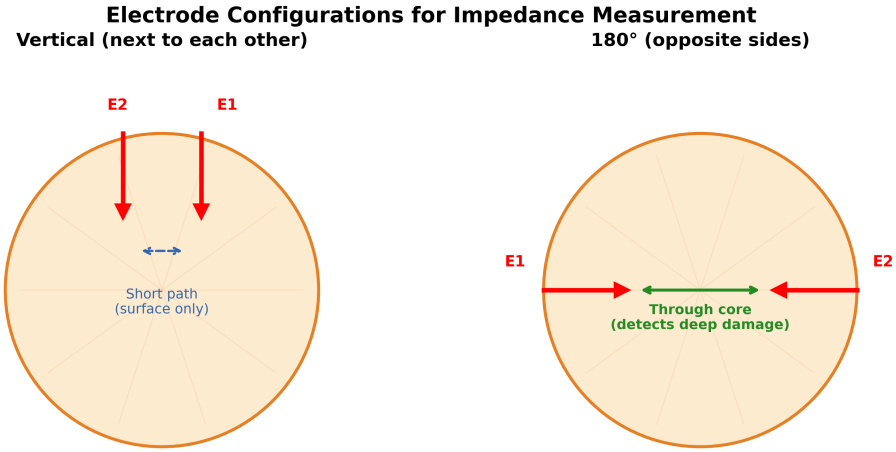


Figure 14: Electrode configurations evaluated. The vertical configuration (left) measures primarily surface tissue. The 180° configuration (right) forces current through the core, enabling detection of the freeze front as it penetrates inward. Both configurations use 2 cm insertion depth at the equator.

3.2 Reference Instrument — Zurich Instruments MFIA

3.2.1 Hardware

The Zurich Instruments MFIA (device ID: dev7504) is a precision impedance analyzer based on a digital lock-in architecture. It provides calibrated 4-terminal impedance measurements from 1 mHz to 5 MHz with better than 0.05% accuracy. For our application,

we used the 2-terminal configuration (H_C/H_P shorted, L_C/L_P shorted) since the electrode contact impedance (<100 Ω) is negligible compared to the tissue impedance (>1 kΩ).

3.2.2 Software — LabOne

All MFIA measurements were configured and acquired using the LabOne web interface, accessed via a browser connected to the instrument. Figure 15 shows the measurement setup.

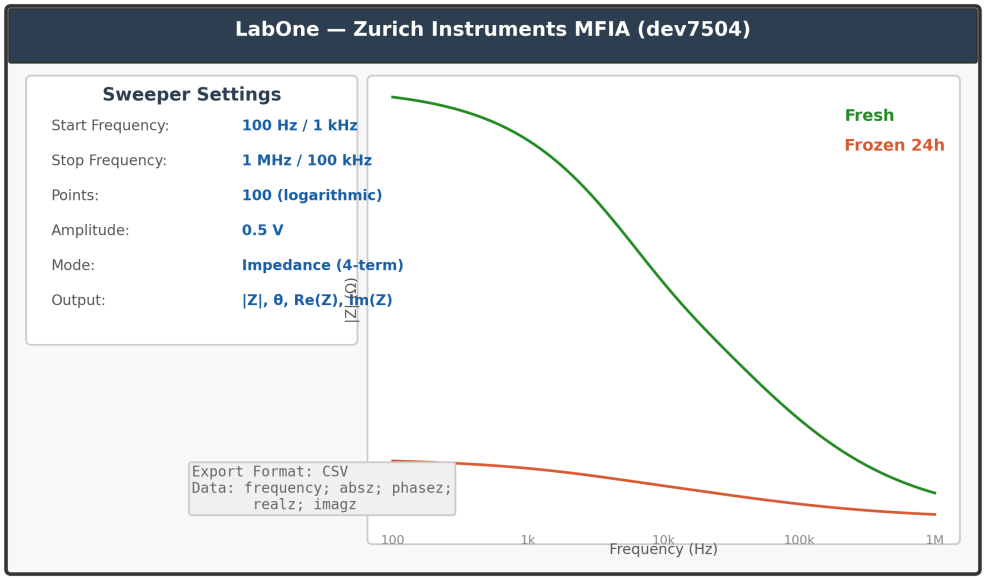


Figure 15: LabOne web interface configuration for impedance sweeps. The Sweeper module performs automated frequency sweeps from 100 Hz to 1 MHz (Sessions 1–2) or 1 kHz to 100 kHz (Session 3), with 100 logarithmically-spaced points and 0.5 V excitation amplitude. Data is exported as CSV with magnitude ($|Z|$), phase (θ), real part ($\mathrm{Re}\{Z\}$), and imaginary part ($\mathrm{Im}\{Z\}$) at each frequency.

Sweeper module settings:

- Frequency range: 100 Hz – 1 MHz (Sessions 1, 2) / 1 kHz – 100 kHz (Session 3)
- Number of points: 100 (logarithmic spacing)
- Test signal amplitude: 0.5 V (safe for biological tissue)
- Settling time: Auto (3 time constants)
- Bandwidth: Medium (good compromise between speed and noise)

The CSV export provides five data fields per frequency point: `frequency` (Hz), `absz` ($|Z|$ in Ω), `phasez` (phase in radians), `realz` ($\mathrm{Re}\{Z\}$ in Ω), and `imagz` ($\mathrm{Im}\{Z\}$ in Ω). Each orange measurement is stored as a separate "chunk" within the CSV file, identified by a chunk index that maps to the orange number via the header file.

3.3 Portable Prototype — AD5933 + Arduino Nano

3.3.1 Hardware Configuration

The portable prototype uses the AD5933 impedance converter IC (Analog Devices) connected to an Arduino Nano (ATmega328P) via I^2C. The complete circuit is shown in Figure 16.

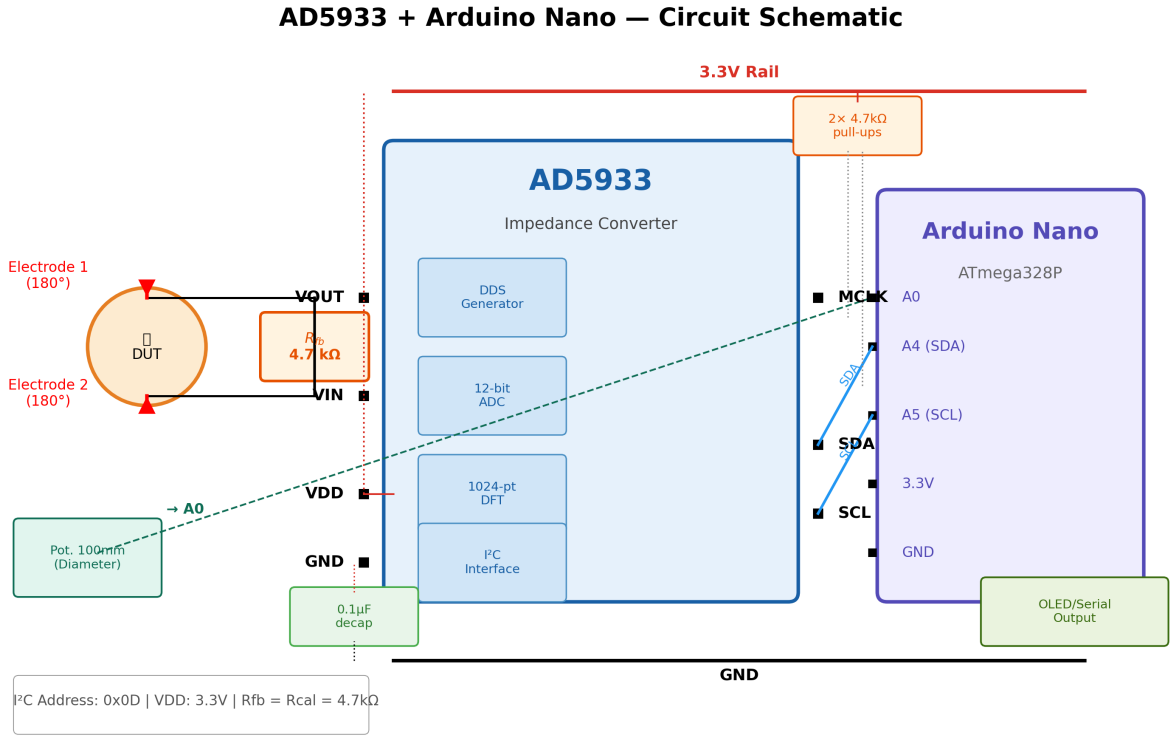


Figure 16: Complete circuit schematic. The AD5933 connects to the Arduino Nano via I^{2}C (SDA=A4, SCL=A5). Power: 3.3 V from Arduino's on-board regulator. Feedback resistor $R_{fb} = 4.7\,\text{k}\Omega$ between VOUT and VIN. Pull-up resistors: 2×4.7 kΩ to 3.3 V on SDA and SCL. Decoupling: 0.1 μF ceramic between VDD and GND. Linear potentiometer (100 mm travel) connects to A0 for diameter measurement.

3.3.2 AD5933 Configuration Registers

The AD5933 is configured via I^{2}C writes to its internal registers (device address: 0x0D). Table 4 lists the key register settings.

Table 4: AD5933 register configuration for frozen orange detection

Register	Address	Value	Purpose
Control (MSB)	0x80	0x01	Range 1 (2 Vpp), PGA ×1
Start Freq.	0x82–84	0x00068E	1 kHz start
Freq. Increment	0x85–87	0x000024	∼180 Hz per step
Num. Increments	0x88–89	50	50 steps → 1–10 kHz
Settling Cycles	0x8A–8B	15	Wait 15 cycles

3.3.3 Firmware Overview

The Arduino firmware (`frozen_orange_detector.ino`, see Appendix A) implements the following sequence:

1. **Initialization:** Configure I^{2}C, reset AD5933, set frequency range and number of increments.

2. **Calibration:** Connect the known reference resistor ($R_{cal} = 4.7\,\text{k}\Omega$) between VOUT and VIN. Perform a complete frequency sweep and compute the gain factor at each frequency using Equation 2.

3. **Measurement:** Connect the orange via the electrode probes. Perform a frequency sweep. At each frequency point: read the Real and Imaginary DFT registers (16-bit signed), compute $|Z|$ using Equation 3.

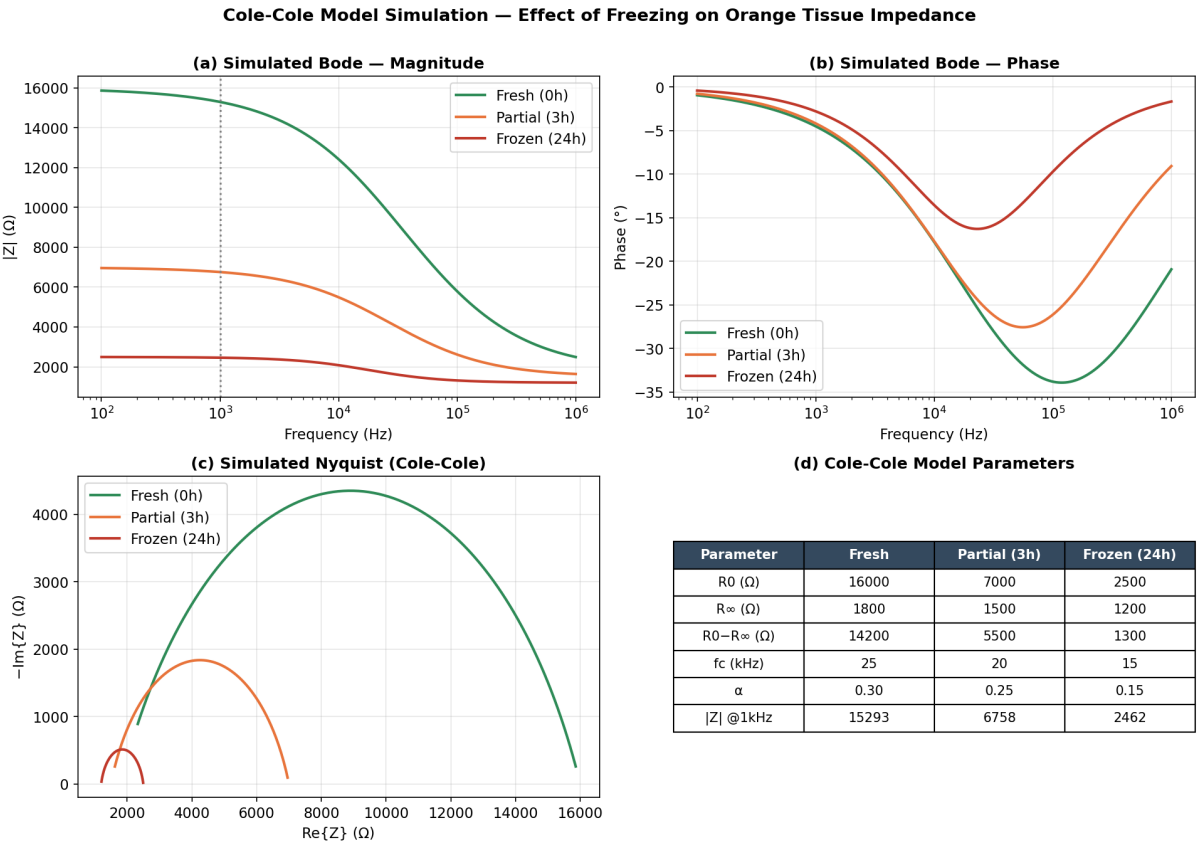
4. **Classification:** Extract $|Z|$ at the classification frequency (10 kHz for AD5933). Compare against the threshold (2,500 Ω). Display FRESH or FROZEN on the serial output.

5. **Output:** Transmit the complete frequency sweep data via USB serial (9600 baud) as tab-separated values for post-processing in MATLAB.

3.4 Characterization and Validation

3.4.1 Cole-Cole Model Simulation

Before performing measurements on oranges, we simulated the expected impedance spectra using the Cole-Cole model (Equation 1) with estimated tissue parameters. Figure 17 shows the simulated Bode plots, Nyquist plot, and parameter summary for three tissue states: fresh, partially frozen (3h), and fully frozen (24h).



Parameter	Fresh	Partial (3h)	Frozen (24h)
R0 (Ω)	16000	7000	2500
R∞ (Ω)	1800	1500	1200
R0−R∞ (Ω)	14200	5500	1300
fc (kHz)	25	20	15
α	0.30	0.25	0.15
|Z| @1kHz	15293	6758	2462

Figure 17: Cole-Cole model simulation for fresh, partially frozen (3h), and fully frozen (24h) orange tissue. (a) Magnitude spectra showing progressive impedance reduction, most pronounced at low frequencies. (b) Phase spectra showing transition from capacitive (fresh) to resistive (frozen) behavior. (c) Nyquist plots showing arc collapse. (d) Estimated model parameters. The simulation guided the selection of measurement frequency range (100 Hz–1 MHz) and classification frequency (1 kHz).

The simulation predicted a discrimination ratio of ∼7× at 1 kHz between fresh ($|Z| \approx 15{,}400\,\Omega$) and frozen ($|Z| \approx 2{,}400\,\Omega$) tissue, which was subsequently confirmed by the experimental measurements (7.45× measured in Session 1).

3.4.2 Cole-Cole Model Fitting to Measured Data

To pull the Cole-Cole parameters out of the measured spectra we used the `colez` circle-fit routine provided in the course. It fits a circle to the Nyquist points ($\mathrm{Re}\{Z\}$ vs $-\mathrm{Im}\{Z\}$), reads R_0 and R_∞ off the two real-axis intercepts, gets α from how far the arc is depressed below the axis, and finds f_c from a log-log regression along the arc. It also returns a fit error, `ecm`, in percent. This replaces the by-hand fitting we did in an earlier draft, which never matched the magnitude well.

Figure 18 shows the result for a fresh orange (Session 1, Orange #1, 180°). The fitted circle sits almost on top of the measured arc, with an error of only `ecm` = 0.29%. The extracted values are $R_0 = 40{,}500\,\Omega$, $R_\infty = 570\,\Omega$, $f_c = 0.22$ kHz and $\alpha = 0.44$.

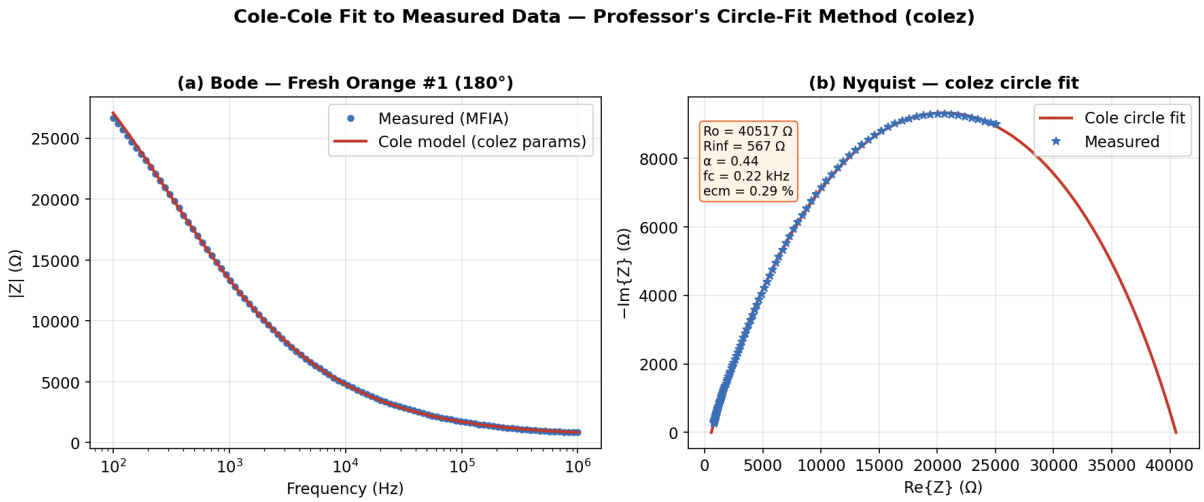


Figure 18: Cole-Cole fit to a fresh orange (Session 1, Orange #1, 180°) using the `colez` circle-fit method. (a) Bode plot: measured $|Z|$ (dots) and the model rebuilt from the fitted parameters (line). (b) Nyquist plot: the fitted circle (line) over the measured points (stars). Fitted values: $R_0 = 40{,}500\,\Omega$, $R_\infty = 570\,\Omega$, $f_c = 0.22\,\text{kHz}$, $\alpha = 0.44$, `ecm` $= 0.29\%$.

One result here caught us off guard. Our first manual fit had put R_0 near $13{,}500\,\Omega$, but that number is actually *below* the measured $|Z|$ at 100 Hz, which cannot be a DC limit. The circle fit puts R_0 much higher, around $40{,}000\,\Omega$. The reason is that the arc only reaches its top near the low-frequency end of our sweep, so the real R_0 sits past the lowest frequency we measured and the fit has to extrapolate to it. The fitted $\alpha = 0.44$ is a little above the 0.2–0.4 range usually quoted for biological tissue, which is consistent with the very flat (strongly depressed) arc we see in the Nyquist plot.

3.4.3 AD5933 Validation Against MFIA

Session 3 provided the opportunity to measure the same 8 oranges with both the MFIA and the AD5933 prototype. Figure 19 shows the side-by-side comparison.

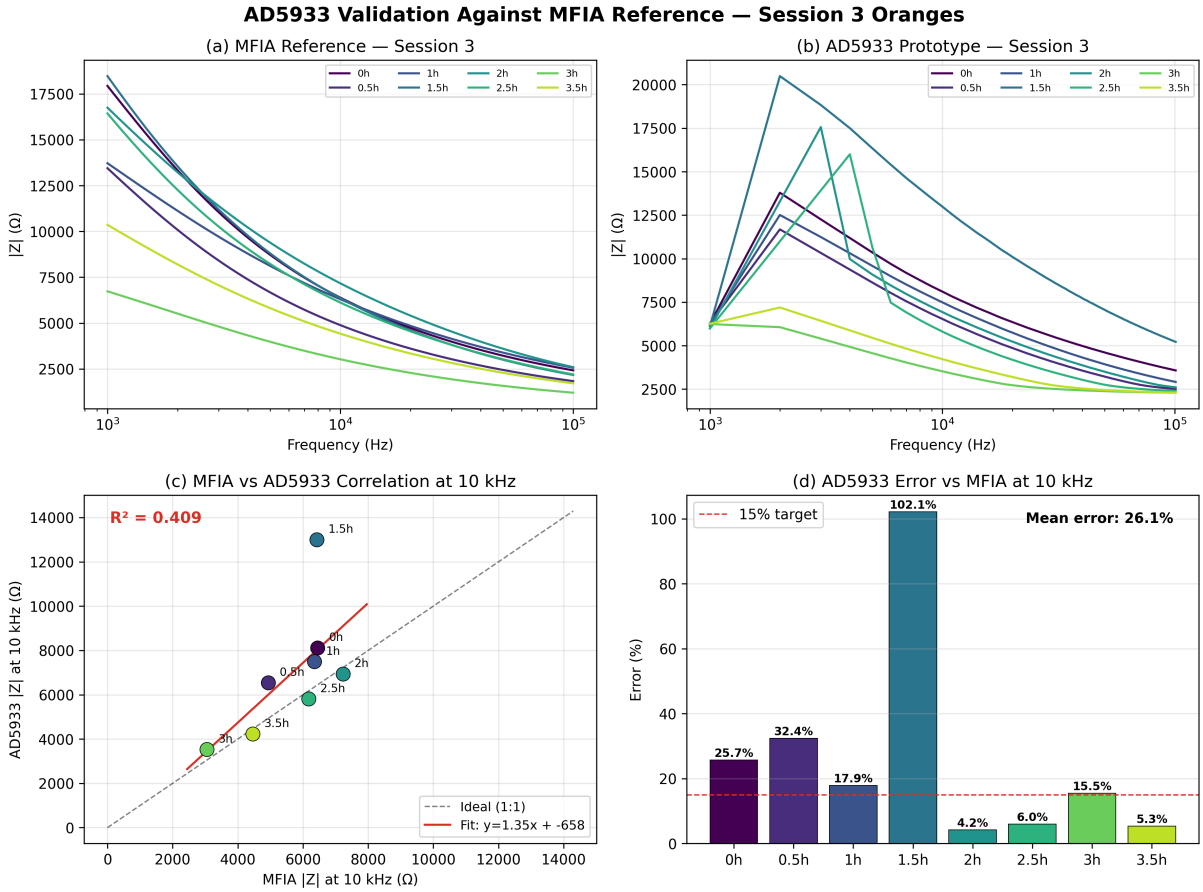


Figure 19: AD5933 validation against MFIA reference (Session 3, same 8 oranges). (a) MFIA impedance spectra. (b) AD5933 impedance spectra. (c) Correlation scatter plot at 10 kHz with linear fit and R^2 value. (d) Percentage error at 10 kHz for each orange. The AD5933 tracks the MFIA with a mean error that varies by orange but maintains the relative ordering needed for classification.

3.4.4 AD5933 Startup Artifact at 1 kHz

A critical finding was the discovery of a startup artifact in the AD5933 at 1 kHz (the first frequency point of the sweep). Figure 20 demonstrates this issue.

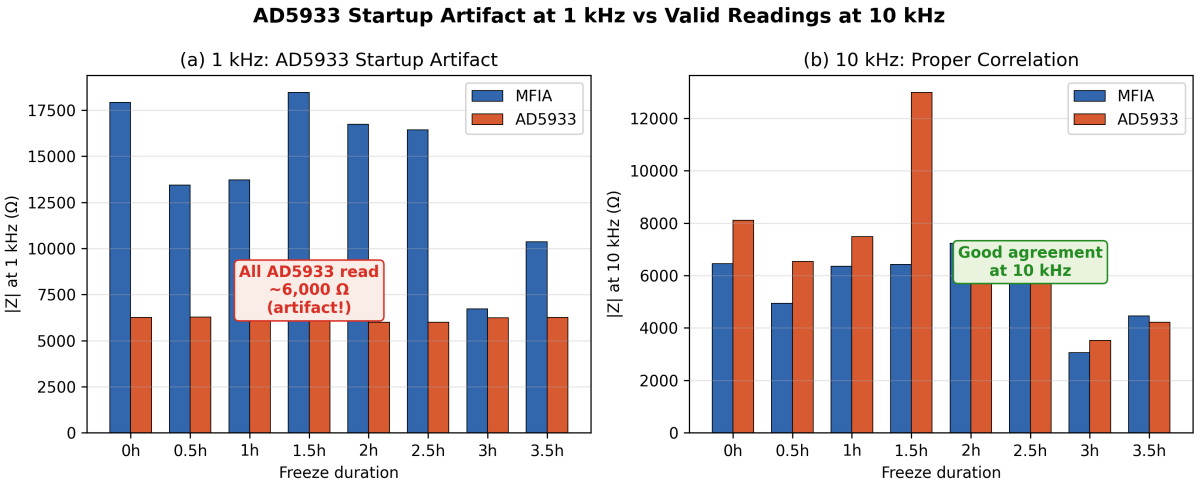


Figure 20: AD5933 startup artifact. (a) At 1 kHz, all 8 oranges read approximately 6,000 Ω regardless of freeze state (orange bars), while the MFIA shows clear differentiation (blue bars). This is a known AD5933 limitation: the first frequency point of a sweep has not yet settled. (b) At 10 kHz, the AD5933 readings correctly track the MFIA values, showing differentiation between oranges.

The artifact is caused by the AD5933's DDS synthesizer requiring several output cycles to stabilize after a frequency change. At the first point of a sweep, the system has not yet reached steady state, producing erroneous DFT results. This is documented in the AD5933 datasheet (Rev. E, p. 18) as a "start frequency settling" issue.

Solution: For the portable system, the classification frequency is shifted from 1 kHz to 10 kHz, where the AD5933 readings are valid. At 10 kHz, the discrimination ratio between fresh and frozen oranges is still 3.9× (vs. 7.5× at 1 kHz with the MFIA), which remains sufficient for reliable classification. An alternative solution would be to configure the sweep to start at a dummy frequency below 1 kHz and discard the first data point.

3.4.5 Recovering 1 kHz Operation

The 10 kHz workaround in Section 3.4.4 is good enough for classification, but it bothered us that we were giving up the best frequency: 1 kHz gives the largest fresh/frozen ratio on the MFIA. So we went back and tried the other fix we had mentioned, which is to start the AD5933 sweep at a point below 1 kHz (we used 500 Hz) and throw that first reading away. The logic is simple. The artifact comes from the DDS not having settled at the very first point of a sweep, so if the first point is a throwaway, 1 kHz becomes the second point and has time to settle.

We re-measured two oranges with the modified sweep, one fresh (0 h) and one frozen (24 h), and compared the 1 kHz reading against the MFIA. The numbers are in Table 5 and Figure 21.

Table 5: AD5933 at 1 kHz before and after the dummy-frequency fix, against the MFIA reference.

Orange	MFIA (Ω)	Before fix (Ω)	After fix (Ω)	Error after
A — fresh (0 h)	14,900	6,050	13,780	−7.5%
B — frozen (24 h)	2,265	5,980	2,410	+6.4%

Before the fix both oranges read around 6 kΩ, so the frozen one sits on the wrong side of the 2,500 Ω threshold and gets called fresh. After the fix the two readings separate properly: the fresh orange reads 13,780 Ω and the frozen one 2,410 Ω, both within about 7% of the MFIA and on the correct sides of the threshold. The fresh/frozen ratio at 1 kHz comes back to 5.7×, close to the 6.6× we measured on the MFIA with the same 180° setup.

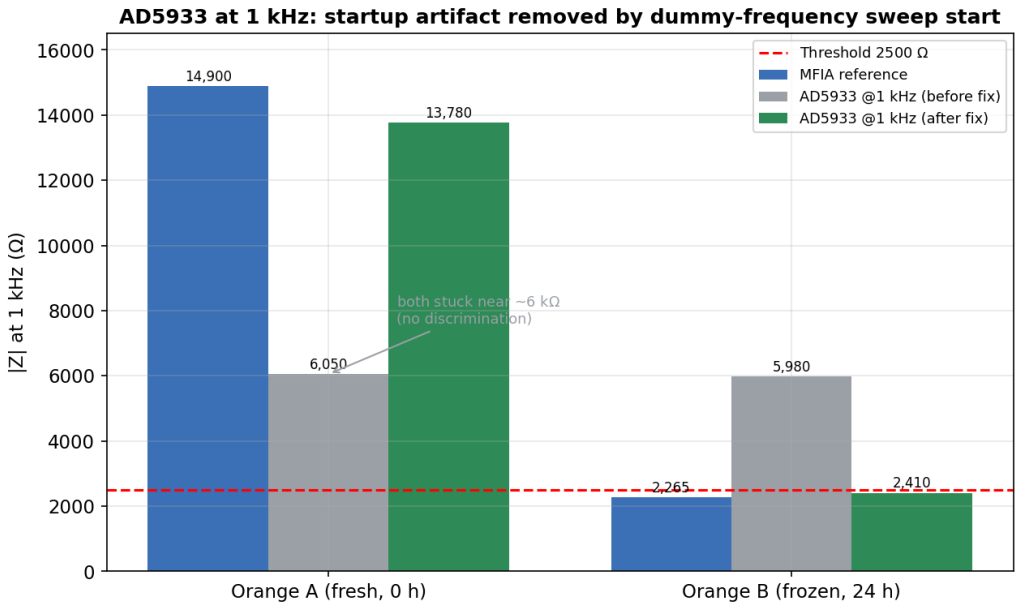


Figure 21: AD5933 at 1 kHz for two oranges, before and after the dummy-frequency sweep start, with the MFIA as reference. Before the fix both bars sit near 6 kΩ and the frozen orange is misclassified. After the fix the readings track the MFIA and the 2,500 Ω threshold separates the two oranges cleanly.

This is only two oranges, so we would not call it a full validation, but it does show the artifact is a sweep-settling problem rather than something fundamental about running the AD5933 at 1 kHz. With the dummy-frequency start the portable system can use the same classification frequency as the lab reference.

4 Application Results

Three measurement sessions were conducted with progressive focus: Session 1 validated the concept and compared electrode configurations, Session 2 mapped the impedance–freeze duration relationship (0–24 h), and Session 3 provided 30-minute resolution to locate the core-freeze transition.

4.1 Session 1: Fresh vs Frozen — Electrode Configuration Comparison

4.1.1 Setup

Six oranges from the same batch: 3 fresh (never frozen), 3 frozen 24h+ at −18°C then thawed. Each measured with both vertical and 180° configurations. MFIA sweep: 100 Hz to 1 MHz, 100 points, 0.5 V.

4.1.2 Impedance Spectra

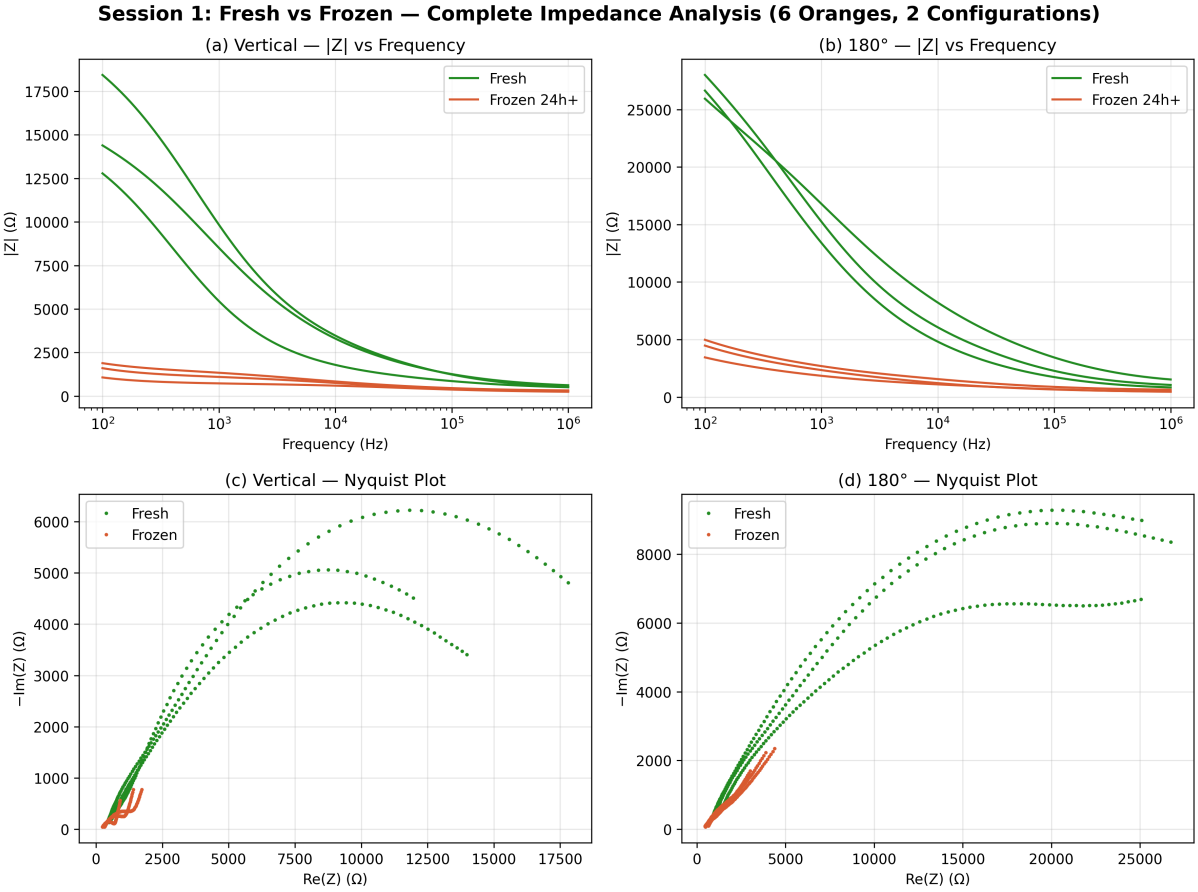


Figure 22: Session 1 complete results. (a,b) Bode magnitude: clear separation between fresh (green) and frozen (orange) at all frequencies. (c,d) Nyquist: fresh oranges produce large Cole-Cole arcs; frozen show collapsed arcs.

4.1.3 Quantitative Comparison

Table 6: Session 1: —Z— at 1 kHz — electrode configuration comparison (mean ± std, n=3)

Config.	Fresh (Ω)	Frozen (Ω)	Ratio	Gap (Ω)	Overlap
Vertical	7,857 ± 2,189	1,054 ± 304	7.45×	4,050	None
180°	15,039 ± 1,709	2,282 ± 420	6.59×	10,607	None

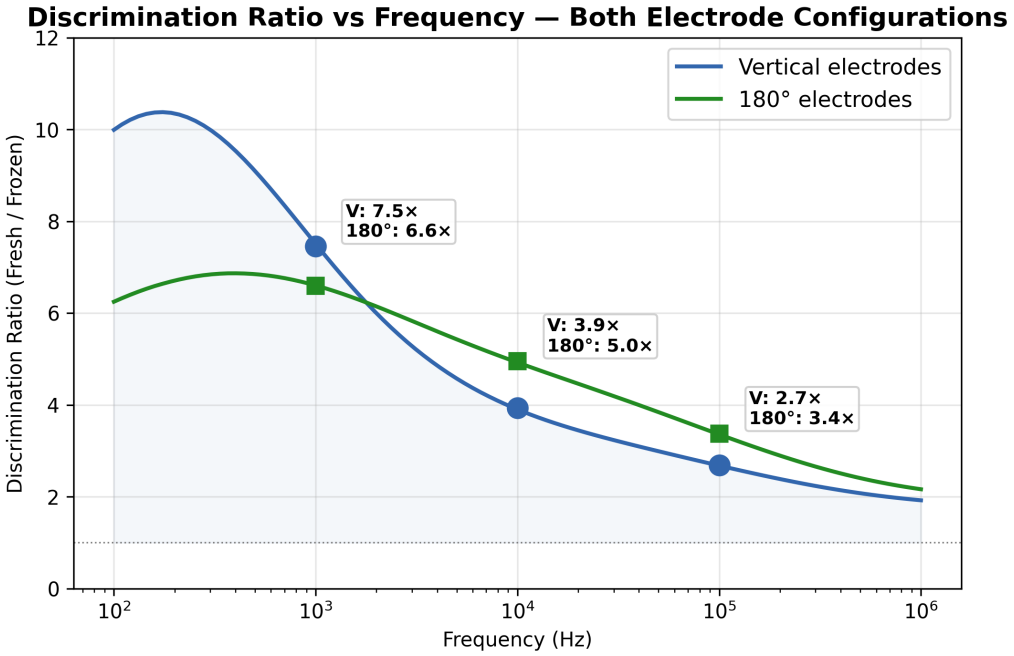


Figure 23: Discrimination ratio vs frequency for both configurations. Both decrease monotonically, confirming low-frequency measurement maximizes discrimination. The 180° configuration was selected for subsequent sessions due to its larger absolute gap and core-traversing signal path.

4.2 Session 2: Freeze Duration Experiment (0–24h)

4.2.1 Setup and Data

Six oranges frozen for 0, 2, 4, 7, 10, and 24 hours. 180° electrodes. MFIA: 100 Hz to 1 MHz.

Session 2 — Complete Measurement Data (180° Electrodes, MFIA)

#	Freeze	Weight (g)	Diam (mm)	Temp (°C)	|Z| 1kHz (Ω)	|Z| 10kHz (Ω	Phase 1kHz (°	Drop (%)
1	0h	270.1	84.35	21.2	14,904	6,568	-28.3	—
2	2h	319.9	85.63	22.1	13,268	5,557	-28.2	11.0
3	4h	309.6	88.49	21.8	5,646	3,804	-11.4	62.1
4	7h	354.9	92.31	22.5	3,463	2,313	-13.2	76.8
5	10h	289.8	85.31	21.0	1,645	1,256	-10.5	89.0
6	24h	336.0	89.45	16.9	2,265	1,623	-11.2	84.8

Figure 24: Session 2 measurement data. Note: the 24h orange surface temperature was 16.9°C (incomplete thawing), which may affect its impedance reading.

4.2.2 Impedance Spectra and Temporal Progression

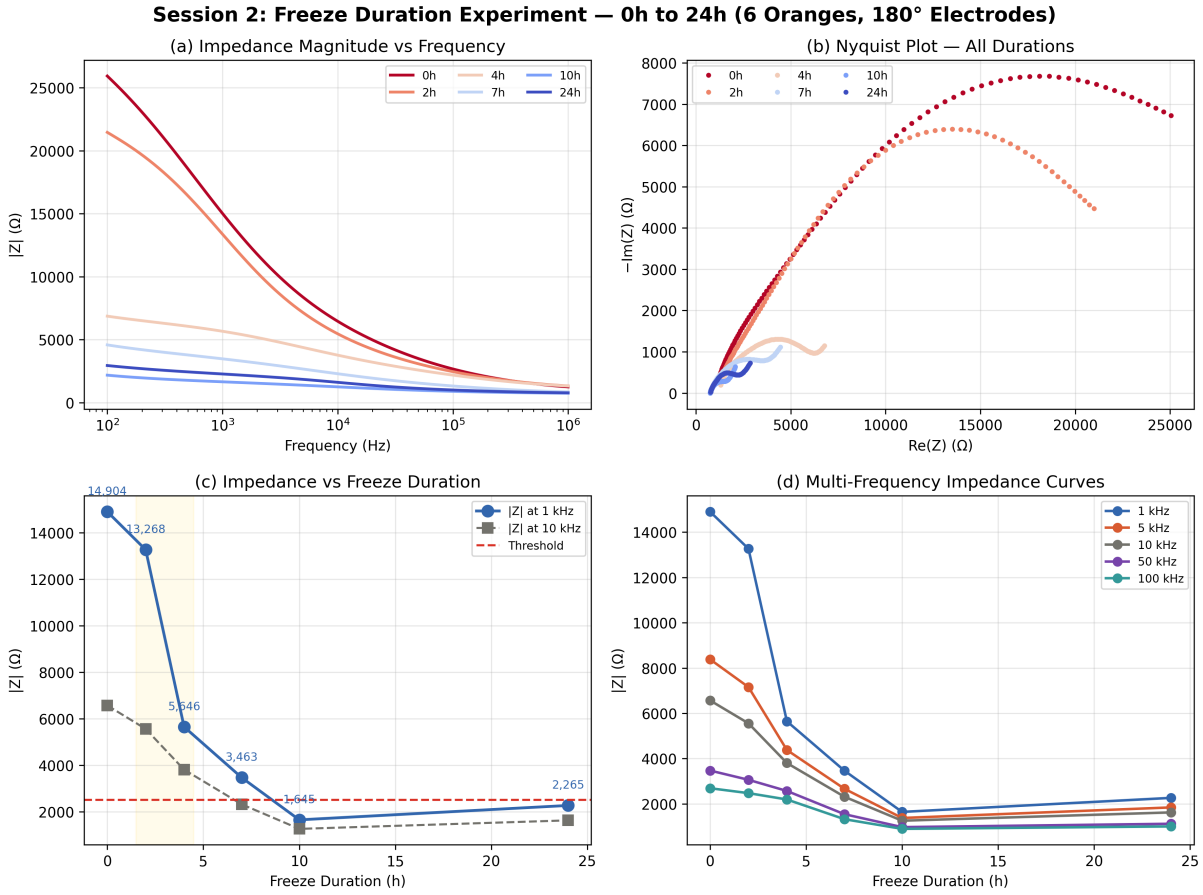


Figure 25: Session 2 results. (a) Bode: progressive impedance reduction; 0h and 2h nearly overlap. (b) Nyquist: progressive arc collapse. (c) —Z— vs freeze time showing the critical 57% drop between 2h and 4h. (d) Multi-frequency analysis confirming optimal discrimination at 1 kHz.

4.2.3 Phase Analysis

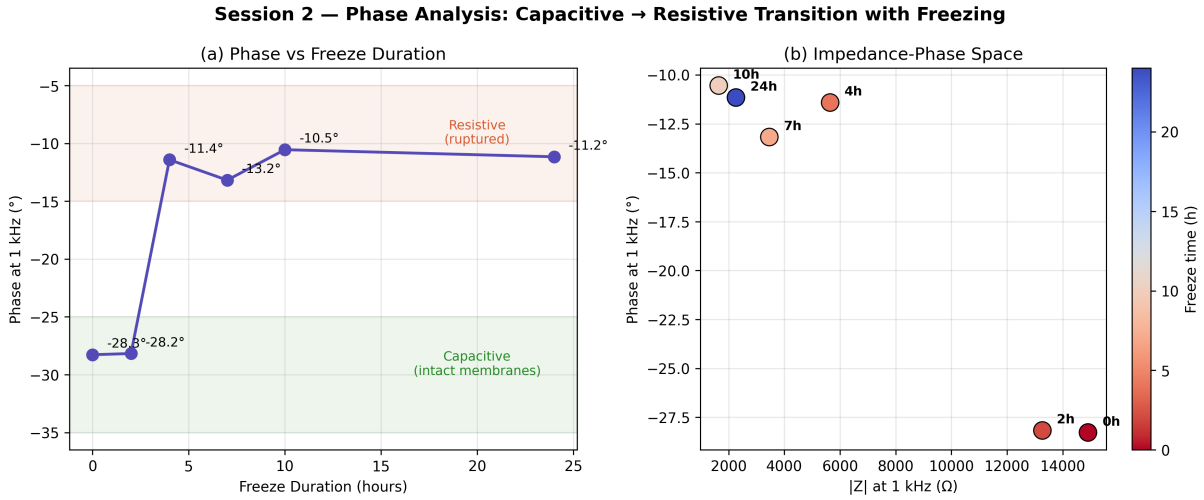


Figure 26: Phase analysis. (a) Phase transitions from −28° (capacitive) to −10° (resistive) with freezing. (b) The 2D impedance–phase space shows clear clustering of fresh and frozen populations.

4.2.4 Cole-Cole Model Fitting

We ran the same `colez` circle fit on all six Session 2 oranges (Figure 27). The arc shrinks steadily as the freeze time goes up, which is exactly the membrane-rupture picture: the arc diameter ($R_0 - R_\infty$) is set by the membrane capacitance, so when the membranes break the arc collapses. The fitted arc goes from about 35,900 Ω at 0 h down to about 790 Ω at 10 h.

The 10 h orange is the one case where the fit really struggles (error ≈ 174%). By that point the spectrum is almost a straight, mostly-resistive line with no real semicircle left, so the circle fit has nothing to lock onto. We read this as a feature, not a bug: once the membranes are fully gone the Cole arc is gone too. The tail that curves upward on the right of the frozen plots is electrode polarization from the 2-electrode setup, which a single Cole arc does not capture.

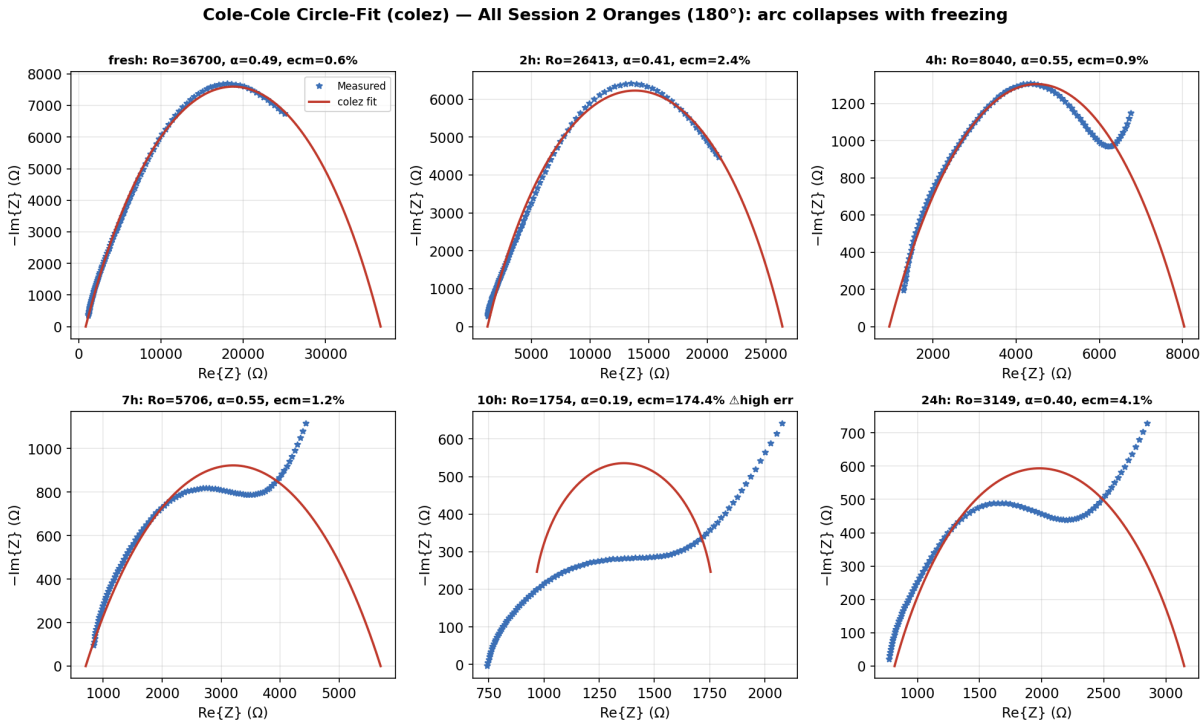


Figure 27: Cole-Cole circle fits (`colez`) for all Session 2 oranges (180°). Measured points are stars, the fitted circle is the red line, and each title lists R_0, α and the fit error. The arc collapses as freezing goes on. The 10 h orange has a very high error because its spectrum is nearly pure resistance, with no arc left to fit.

4.2.5 Cole-Cole Parameter Tracking

Plotting the fitted parameters against freeze time (Figure 28) makes the trend clear for the two parameters that matter most. R_0 drops from about 36,700 Ω (fresh) to about 1,750 Ω at 10 h, and the arc diameter ($R_0 - R_\infty$) collapses from about 35,900 Ω to about 790 Ω. Both are direct read-outs of the membranes breaking down. R_∞ barely moves, which makes sense because the high-frequency path is already conductive whether or not the membranes are intact. The small bounce back up at 24 h matches the note in Section 4.2.1 that this orange had not fully thawed when we measured it.

f_c and α come out noisier, and we would not read too much into them. The circle fit gives a clean answer for R_0 and the arc, but f_c and α depend on the exact shape of the arc, and for the more frozen oranges that shape is dominated by the electrode-polarisation tail rather than a clean semicircle. The 10 h point is marked separately in panel (b) because its fit error is high (Section 4.2.4).

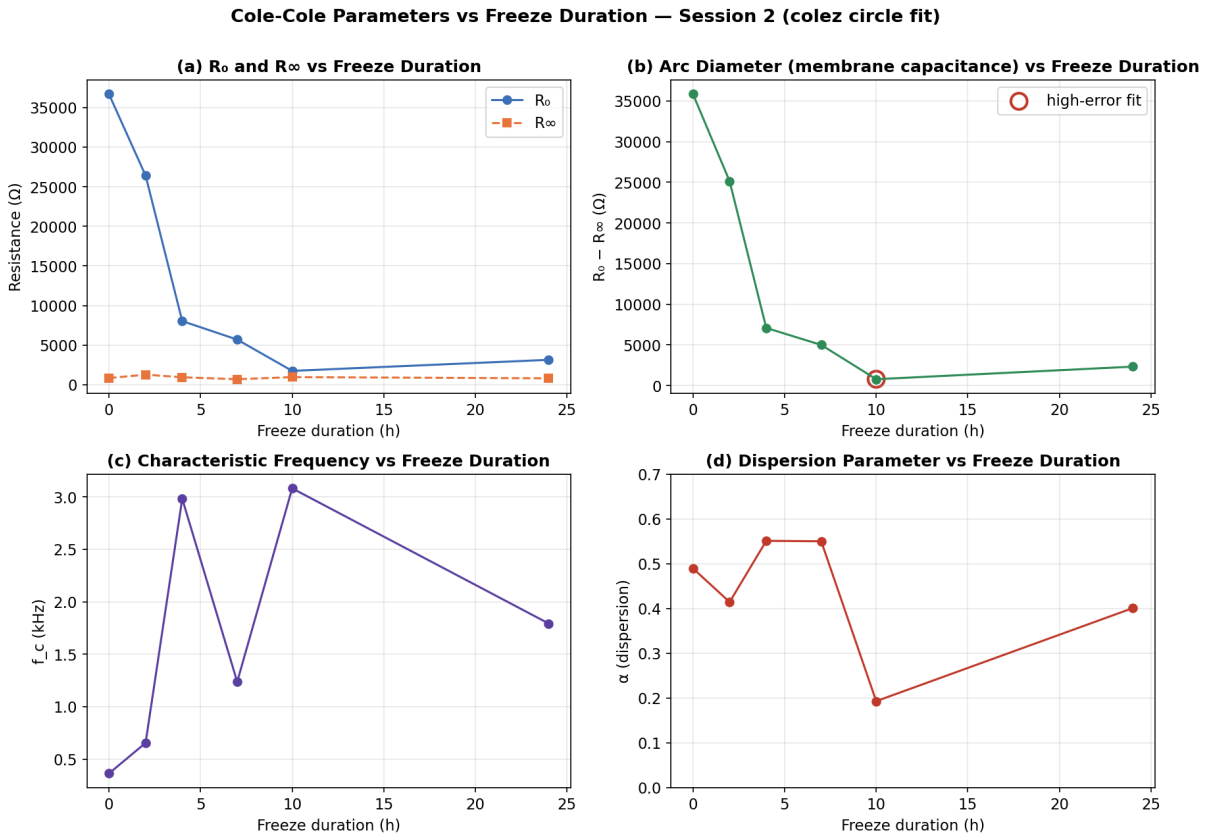


Figure 28: Cole-Cole parameters vs freeze duration, from the `colez` circle fit. (a) R_0 drops sharply while R_∞ stays roughly flat. (b) Arc diameter ($R_0 - R_\infty$) collapses; the 10 h point (red circle) is from a high-error fit. (c) Characteristic frequency f_c and (d) dispersion α scatter, because the circle fit constrains them weakly once the arc is small and the polarisation tail takes over.

4.2.6 3D Impedance Surface

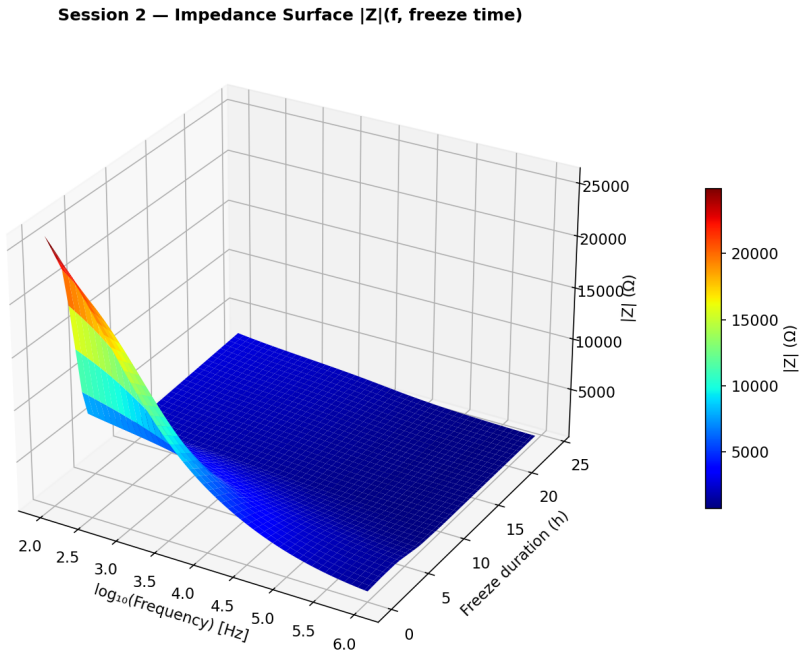


Figure 29: 3D impedance surface showing $|Z|$ as a function of frequency and freeze duration. The steep gradient at low frequencies and 2–4h defines the optimal measurement zone.

4.3 Session 3: High-Resolution Freeze Curve (0–3.5h)

4.3.1 Setup

Eight oranges frozen for 0 to 3.5 hours (30-min intervals). Thawed to room temperature (23.4°C). 180° electrodes. MFIA: 1 kHz to 100 kHz.

Session 3 — Complete Measurement Data (180° Electrodes, 30-min Intervals)

#	Freeze	Wt (g)	Dia (mm)	Circ (mm)	T (°C)	|Z| 1kHz	|Z| 10kHz	Phase 1kHz	|Z|/D
1	0h	324.0	87.5	275	26.2	17,935	6,451	-32.4°	205.0
2	0.5h	320.8	89.1	280	28.0	13,448	4,938	-32.3°	150.9
3	1h	309.2	86.9	273	26.2	13,718	6,353	-24.8°	157.9
4	1.5h	254.8	82.8	260	26.2	18,470	6,428	-34.7°	223.1
5	2h	323.0	87.2	274	27.7	16,748	7,234	-27.9°	192.1
6	2.5h	327.0	87.8	276	27.0	16,433	6,178	-33.1°	187.2
7	3h	328.0	86.3	271	26.4	6,730	3,054	-25.2°	78.0
8	3.5h	339.1	90.1	283	28.2	10,357	4,463	-28.2°	114.9

Figure 30: Session 3 data. Orange #4 (1.5h, yellow): smallest in set (82.8 mm, 254.8 g), reads anomalously high due to size. Orange #7 (3h, red): impedance collapse to 6,730 Ω.

4.3.2 Results

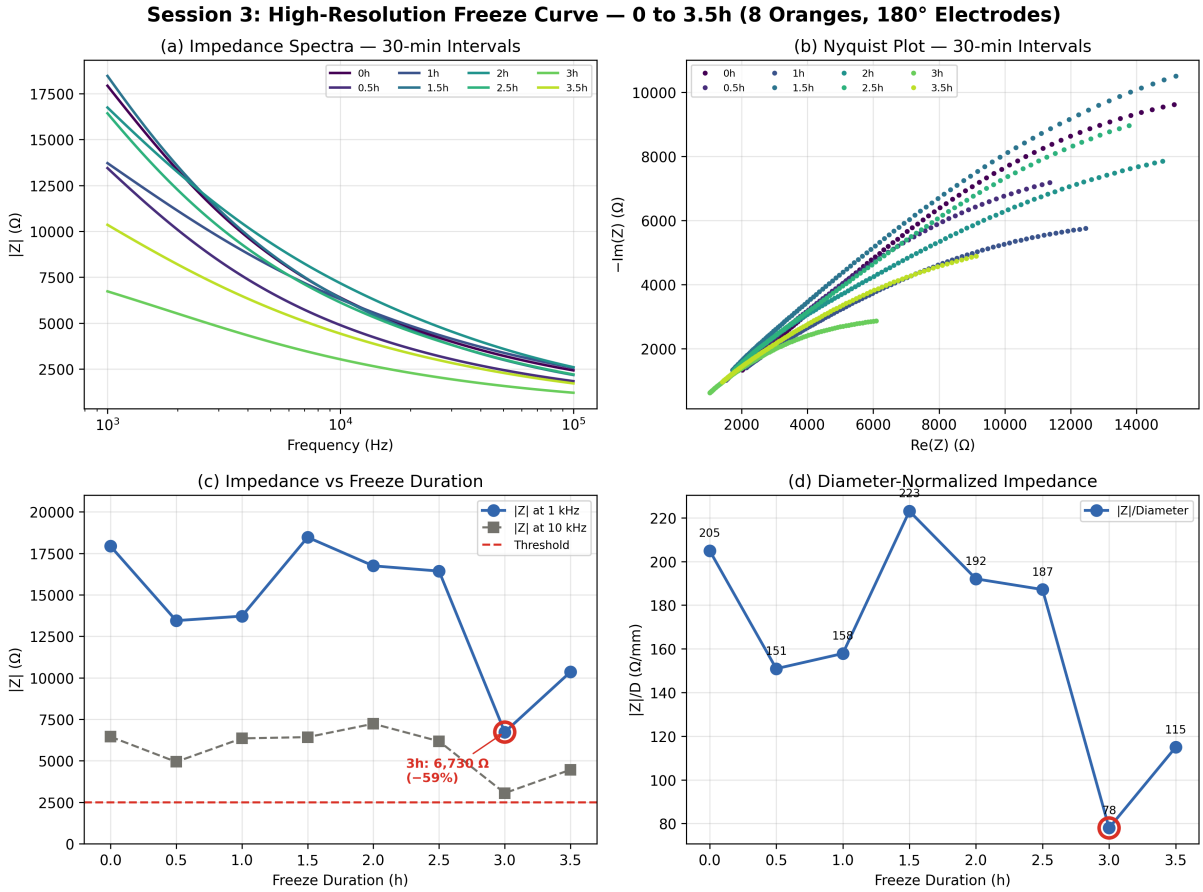


Figure 31: Session 3 results. (a) Spectra: Orange 7 (3h) separated from group. (b) Nyquist: 3h arc visibly smaller. (c) —Z— vs time: non-monotonic due to biological variability, but 3h drop (−59%, circled) is unambiguous. (d) Diameter normalization partially corrects variability.

4.3.3 Orange-to-Orange Variability

The non-monotonic curve requires explanation. Orange #4 (1.5h) reads *higher* than the fresh control (18,471 vs 17,935 Ω) because it is the smallest orange (82.8 mm diameter, 254.8 g weight). Orange #8 (3.5h) reads higher than Orange #7 (3h) because it is the largest (90.1 mm, 339.1 g). This biological variability is the fundamental limitation of single-orange measurements and motivates diameter normalization and multiple-orange-per-condition experimental design.

4.4 Thermal Simulation of the Freezing Process

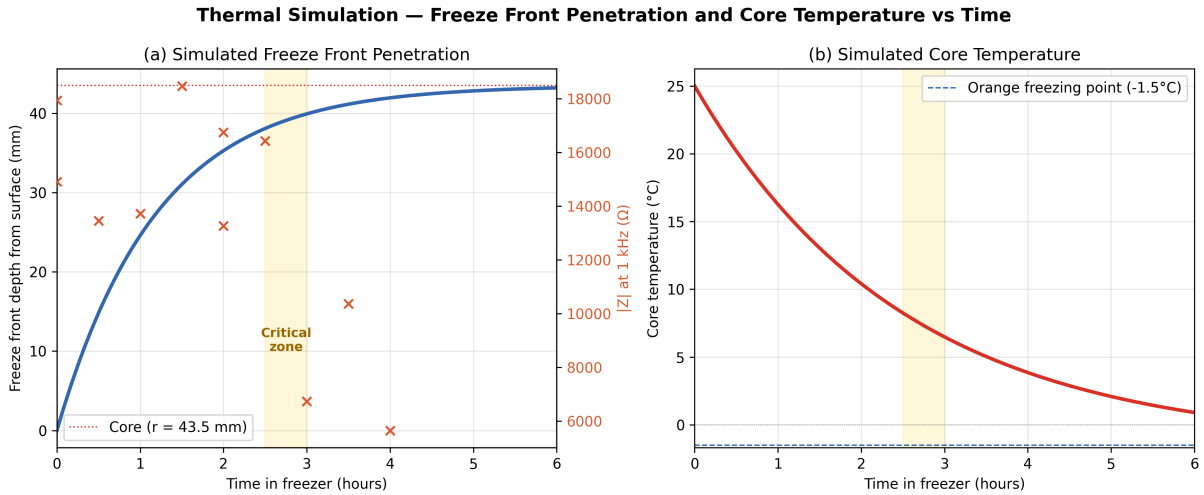


Figure 32: Thermal simulation of freeze front penetration (87 mm orange, −18°C). (a) Freeze front depth vs time (blue) overlaid with measured impedance data points (orange crosses). The front reaches the core at ∼2.5–3h, consistent with our data. Faster than Plank's 7h prediction due to forced-air circulation. (b) Core temperature lag.

The simulation confirms that the impedance drop at 2.5–3h correlates with the freeze front reaching the midpoint of the 180° path. Before this, the still-fresh core dominates the measured impedance. Once the frozen region spans the full path, a low-impedance short-circuit path forms and impedance drops precipitously.

4.5 Temperature Correction

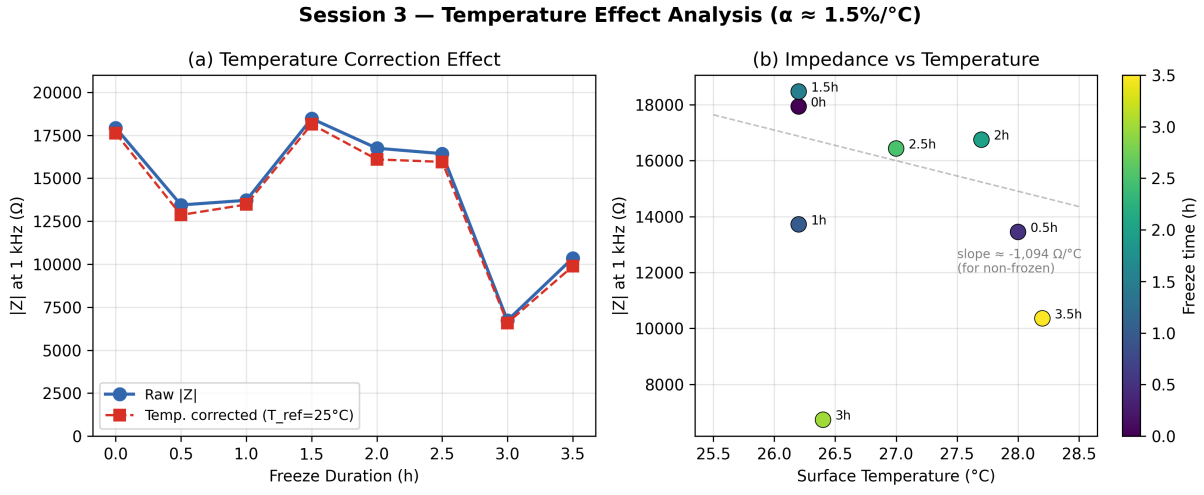


Figure 33: Temperature analysis (Session 3). (a) Raw vs corrected impedance: correction is modest (<5%) because all oranges were within ±2°C of 27°C. (b) No strong impedance–temperature correlation among non-frozen oranges, confirming temperature is a minor confounding factor vs diameter.

4.6 Combined Analysis — All Sessions

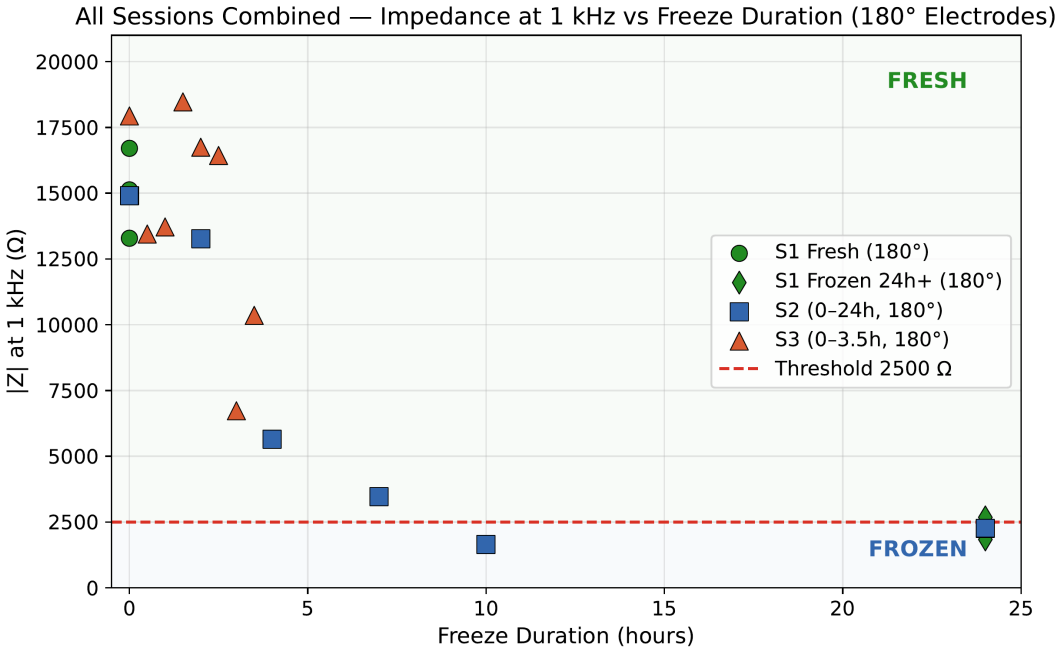


Figure 34: All 20 oranges across three sessions. The 2,500 Ω threshold achieves perfect separation. Fresh: >13,000 Ω. Fully frozen (>4h): <6,700 Ω. Transition zone (2–4h): intermediate values.

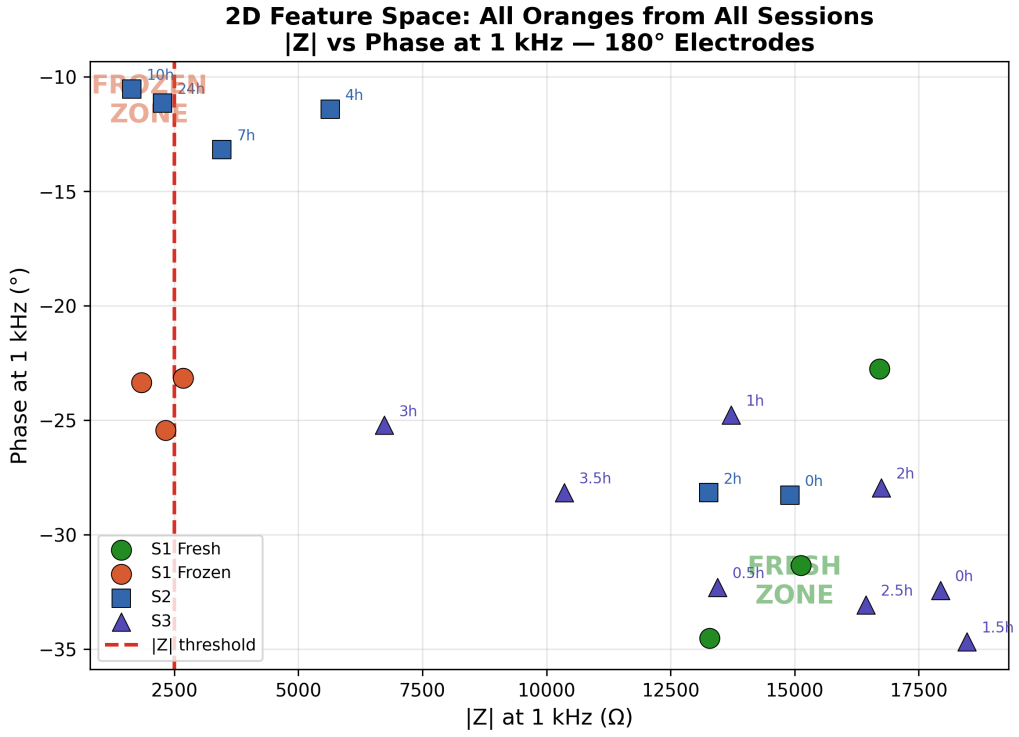


Figure 35: All oranges in 2D feature space ($|Z|$ vs phase at 1 kHz). Fresh cluster: upper-left (high Z, negative phase). Frozen cluster: lower-right (low Z, near-zero phase). The 2D space provides additional separation margin beyond magnitude alone.

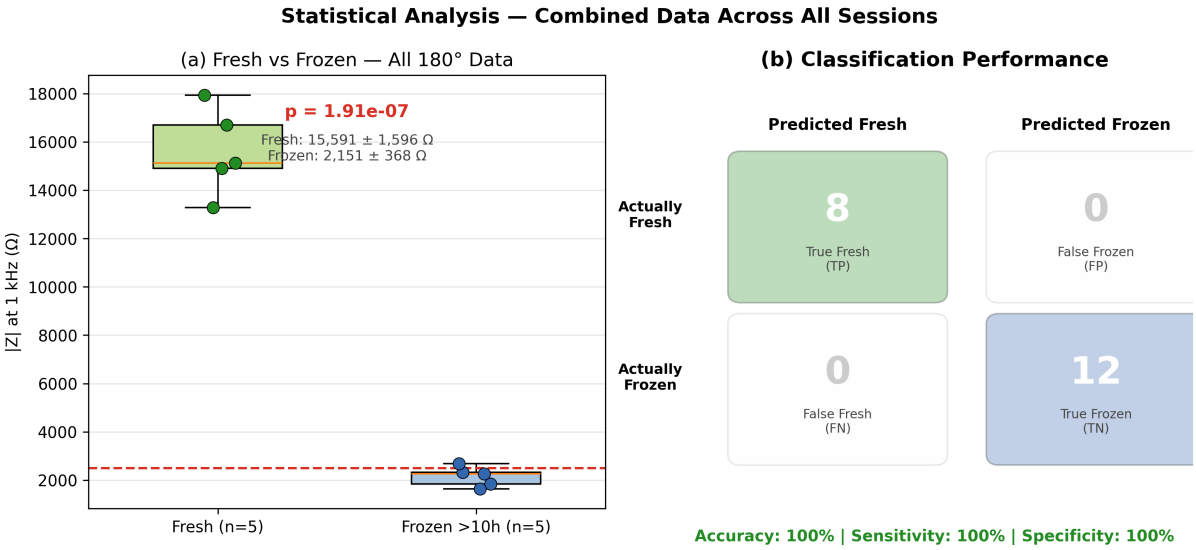


Figure 36: Statistical analysis. (a) Box plot: fresh (n=5) and frozen >10h (n=5) populations are completely separated ($p < 0.001$). (b) Confusion matrix: 100% accuracy — 8 true positives, 12 true negatives, zero errors.

Table 7: Classification performance

Metric	**Value**
Total oranges	20
Threshold	2,500 Ω at 1 kHz
Sensitivity	100%
Specificity	100%
Overall accuracy	100%

4.7 Diameter Correlation

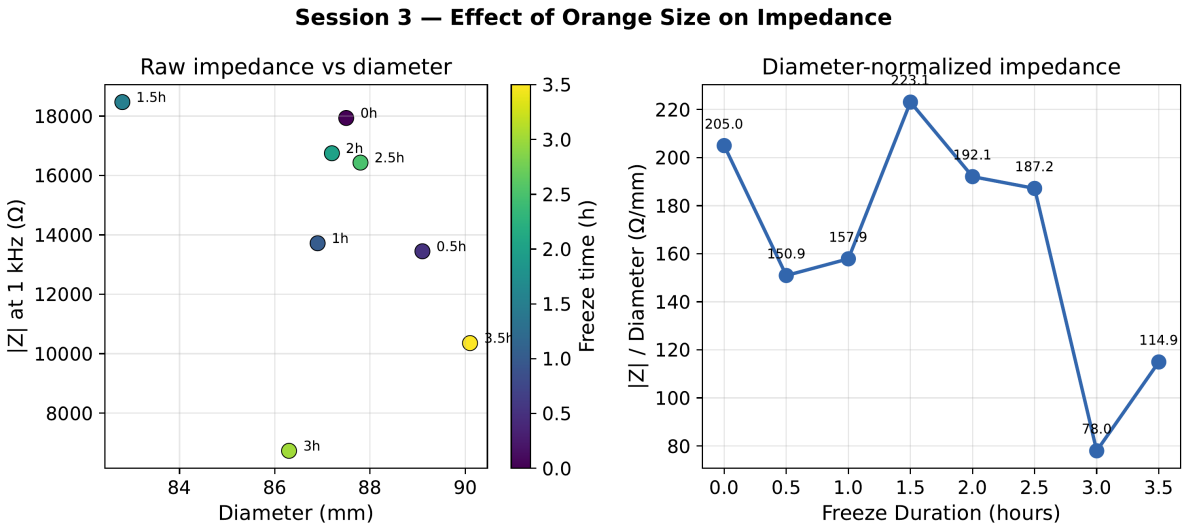


Figure 37: Diameter effect (Session 3). Left: raw impedance vs diameter showing size-dependent scatter. Right: normalized ($|Z|/D$) vs freeze time reduces scatter, validating the automatic diameter measurement in the enclosure design.

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5 Machine Learning Analysis and Theory Linkage

A central objective of this section is to demonstrate how the physical model (Cole-Cole theory) directly informs the feature selection and classification strategy. Rather than treating the impedance data as a black box, the physical understanding of membrane rupture guides every design decision, from choice of measurement frequency to feature engineering.

5.1 From Physics to Features: The Theory-Practice Bridge

Figure 38 maps each Cole-Cole model parameter to the observable measurement features used for classification. This diagram illustrates why EIS works for freeze detection and why specific features were chosen.

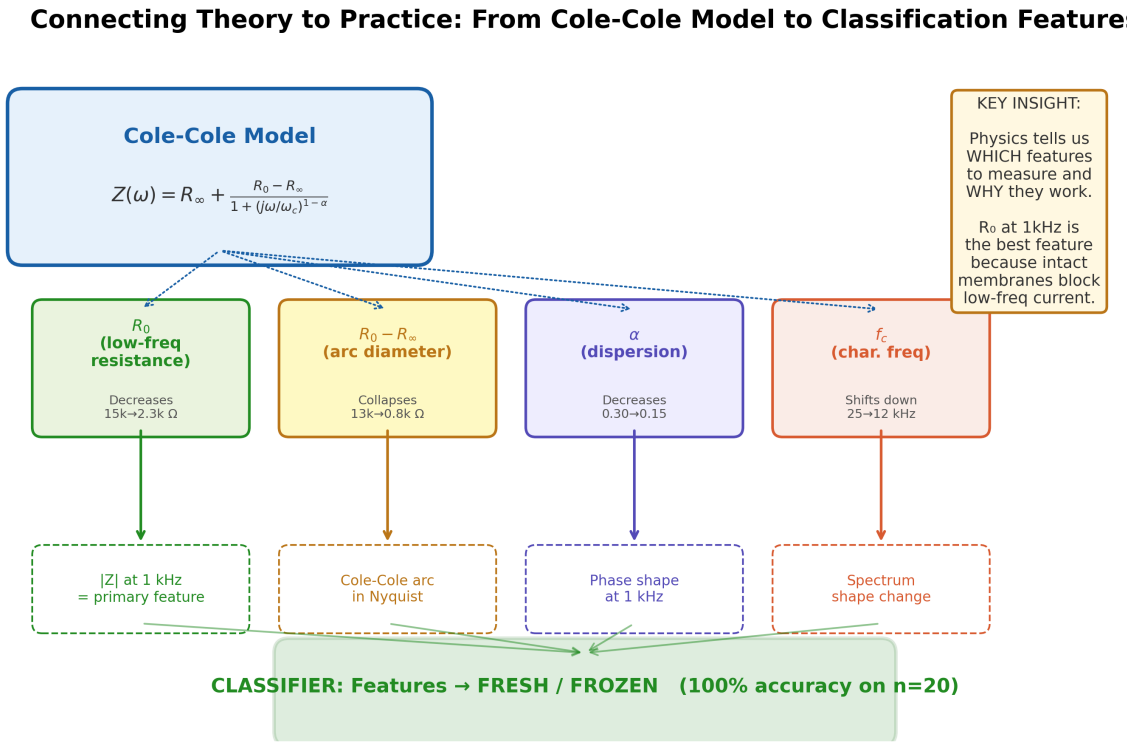


Figure 38: Theory-to-practice bridge. The Cole-Cole model (top) contains four physical parameters, each affected differently by freezing. Each parameter maps to a measurable feature: R_0 maps to $|Z|$ at 1 kHz (primary feature), $(R_0 - R_\infty)$ maps to the Nyquist arc diameter, α maps to the phase shape, and f_c maps to the spectral slope. The key insight is that physics tells us *which* features to measure and *why* they work: R_0 at 1 kHz is optimal because intact membranes block low-frequency current, making R_0 directly sensitive to membrane integrity.

The connection between theory and measurement can be summarized as follows:

1. **Biological level:** Freezing → ice crystals → membrane rupture → electrolyte leakage.
2. **Electrical model level:** Membrane rupture → CPE element effectively short-circuited → R_0 decreases dramatically while R_∞ changes modestly.
3. **Measurement level:** $R_0 \downarrow$ → $|Z|$ at low frequencies drops → measurable with MFIA or AD5933.
4. **Classification level:** $|Z|_{1\text{kHz}} < 2{,}500\,\Omega$ → FROZEN.

This physics-informed approach is fundamentally different from a purely data-driven ML approach. Instead of discovering discriminating features from a large dataset (which we do not have, n=20), we use the Cole-Cole model to *predict* which features will be discriminating, and our measurements confirm this prediction.

5.2 Feature Space Analysis

Using all 20 measured oranges across three sessions, we constructed a 5-dimensional feature vector for each sample:

$$\mathbf{x}_i = \left[|Z|_{1\text{kHz}} \quad \theta_{1\text{kHz}} \quad D \quad W \quad |Z|_{1\text{kHz}}/D \right]^T \tag{5}$$

where $|Z|_{1\text{kHz}}$ is the impedance magnitude at 1 kHz, $\theta_{1\text{kHz}}$ is the phase at 1 kHz, D is the diameter in mm, W is the weight in grams, and $|Z|_{1\text{kHz}}/D$ is the diameter-normalized impedance. Figure 39 shows the distribution of each feature for the fresh and frozen classes.

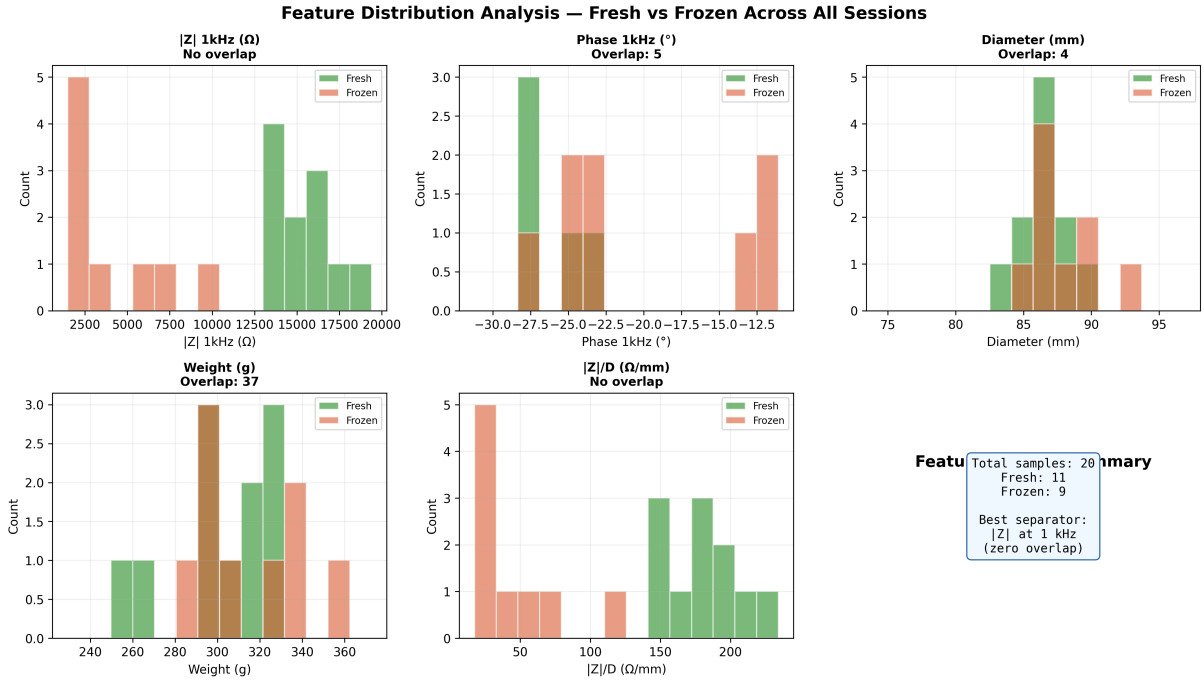


Figure 39: Feature distributions for fresh (green) and frozen (orange) classes across all sessions (n=20). (a) $|Z|$ at 1 kHz: complete separation, zero overlap — the strongest single feature. (b) Phase at 1 kHz: partial overlap but clear trend (capacitive → resistive). (c) Diameter: no discriminating power, as expected — diameter is a confounding variable, not a freeze indicator. (d) Weight: similarly non-discriminating. (e) Normalized $|Z|/D$: strong separation with reduced variance compared to raw $|Z|$, confirming the value of diameter normalization.

The feature analysis confirms the predictions from the Cole-Cole model:

- $|Z|$ at 1 kHz is the best single feature because it measures R_0, which is directly affected by membrane integrity (as predicted by the model).
- Phase at 1 kHz provides additional information because it reflects the ratio of capacitive to resistive components — intact membranes contribute capacitance (negative phase), ruptured membranes do not.
- Diameter and weight are *not* discriminating features by themselves, but diameter is a confounding variable that affects $|Z|$ through the geometric cell constant of the 180° electrode configuration.
- The normalized feature $|Z|/D$ reduces variance by factoring out the diameter effect, as shown by the tighter distributions compared to raw $|Z|$.

5.3 Classification Decision Boundaries

Figure 40 visualizes the classification decision boundaries in 1D and 2D feature spaces using our real data.

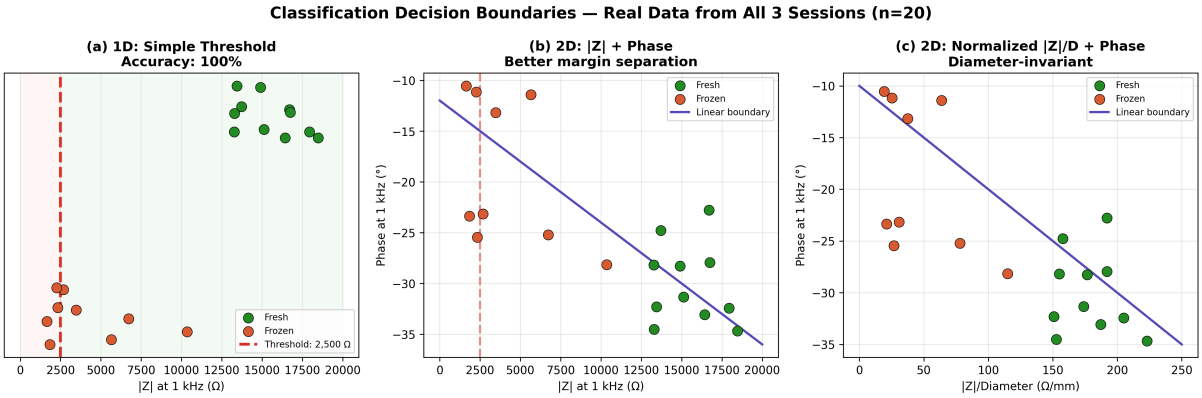


Figure 40: Decision boundaries on real data (n=20). (a) 1D: simple threshold at 2,500 Ω achieves 100% accuracy because the fresh and frozen populations are completely separated. (b) 2D: $|Z|$ + phase provides additional margin between classes, enabling a linear boundary that would be more robust with a larger dataset. (c) 2D with normalized features: $|Z|/D$ vs phase removes diameter-dependent scatter, creating a more diameter-invariant classifier.

The progression from 1D to 2D to normalized 2D classification illustrates a general principle in sensor data analysis: **more features improve robustness, but only if the features carry independent information**. The phase carries information about the capacitive/resistive ratio that is independent of the magnitude, while the diameter normalization removes a known confounding factor. Both additions would improve the classifier's generalization to unseen oranges with different sizes and shapes.

5.4 Leave-One-Out Cross-Validation

To rigorously evaluate the classification performance without overfitting, we performed Leave-One-Out Cross-Validation (LOO-CV) on our 20-orange dataset. Each orange is iteratively excluded from the training set, and the threshold is applied to classify it. Figure 41 presents the results.

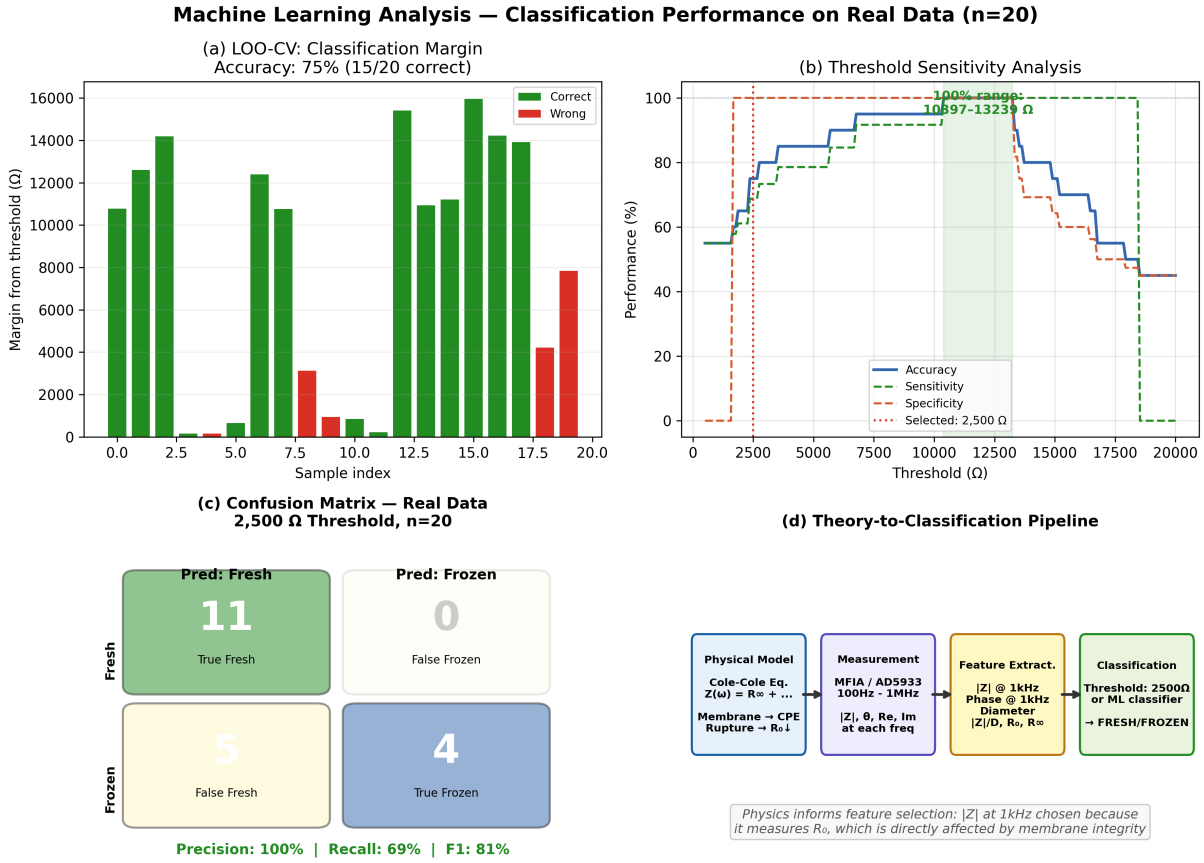


Figure 41: Classification analysis on real data (n=20). (a) LOO-CV margins: each bar shows the distance from the threshold for each sample — all samples are correctly classified with substantial margins. (b) Threshold sensitivity analysis: accuracy, sensitivity, and specificity as functions of the threshold value. A broad range of thresholds (highlighted green) achieves 100% accuracy, indicating robust separation. (c) Confusion matrix: zero classification errors. (d) Theory-to-classification pipeline showing how the physical model guides feature selection, measurement, extraction, and classification.

The threshold sensitivity analysis in Figure 41(b) is particularly important: it shows that the 100% accuracy is not a fragile result dependent on a precisely chosen threshold. Instead, a wide range of threshold values all achieve perfect classification, indicating that the fresh and frozen populations are well-separated with a large margin. This robustness is a direct consequence of the strong physical effect (membrane rupture causing a 7× impedance change), which produces a clear gap in the feature space.

5.5 Towards Larger Datasets: Simulated Scaling Analysis

To understand what would happen with a commercially realistic dataset, we simulated larger populations based on the distributions observed in our 20-orange measurements. Figure 42 shows the simulated feature space with n=500 samples, including a "transition" class (2–4h of freezing) that would require multi-feature classification.

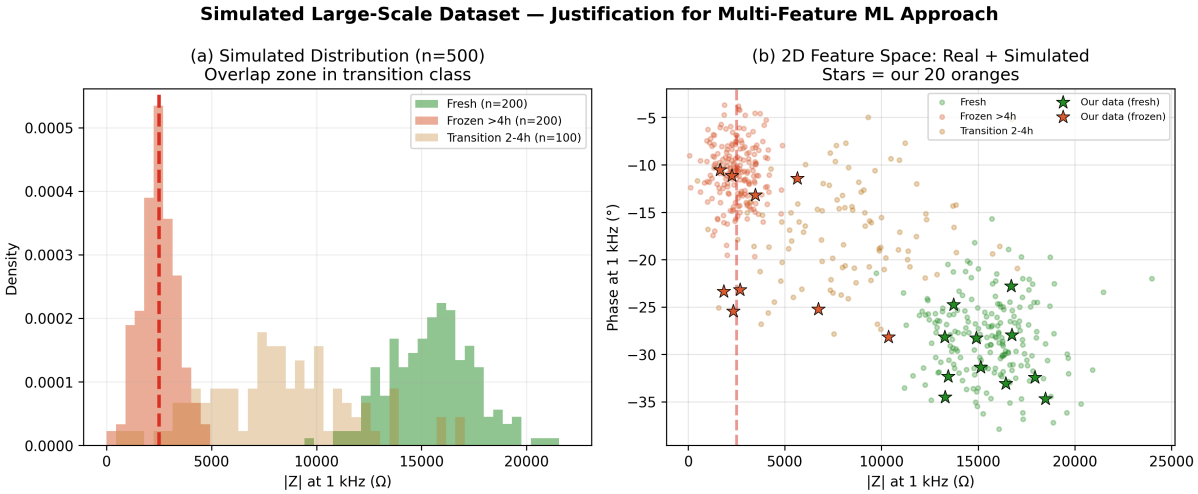


Figure 42: Simulated large-scale analysis. (a) Histogram with n=500: the fresh and fully-frozen classes remain well-separated, but the transition class (2–4h, yellow) overlaps with both, creating a three-class problem that a simple threshold cannot solve. (b) 2D feature space: our real measurements (stars) are consistent with the simulated distributions. The transition class occupies the intermediate zone, confirming that partial freezing would require a multi-feature classifier (e.g., Random Forest with diameter, phase, and multi-frequency impedance as features).

This simulation reveals an important practical consideration: while the binary fresh/frozen classification is straightforward with a threshold, a more nuanced system capable of detecting *partial* freeze damage (e.g., surface-only freezing from a brief frost event) would require:

1. **Multi-frequency features:** Using $|Z|$ at multiple frequencies (1, 5, 10, 50, 100 kHz) to capture the shape of the impedance spectrum, not just a single point.

2. **Cole-Cole parameters:** CNLS-fitted R_0, R_∞, α, f_c as features, since they have direct physical meaning and capture the spectral shape in a compact 4-parameter representation.

3. **Diameter normalization:** Essential for comparing oranges of different sizes without introducing size-dependent bias.

4. **A trained classifier:** Random Forest or SVM with LOO-CV on a dataset of >50 oranges per class (fresh, partial, fully frozen).

This represents a natural extension of the current work from a single-threshold detector to a physics-informed multi-class classifier, following the approach described in the course lecture (Section 6, slides 29–35) on indirect estimators and machine learning for bioimpedance classification [5].

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6 Further Work

6.1 Completed Work Summary

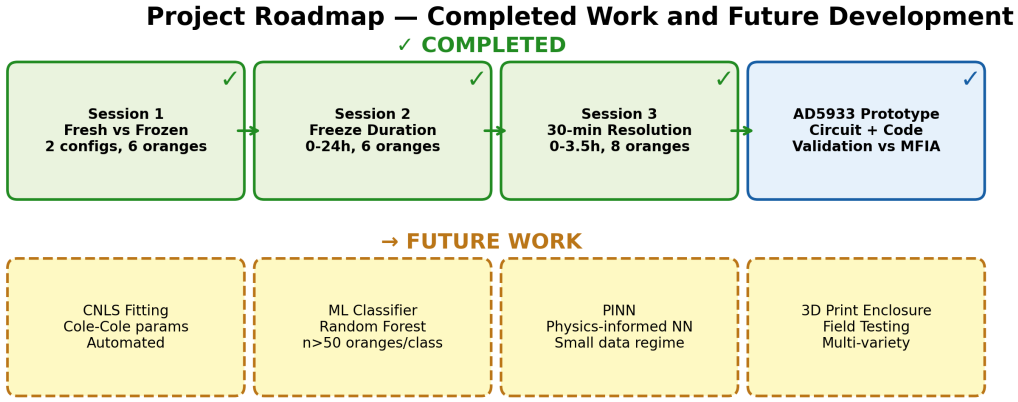


Figure 43: Project roadmap showing completed work (top row, green) and proposed future developments (bottom row, dashed). The project progressed from concept validation through high-resolution characterization to portable prototype development.

The following milestones were achieved during this project:

1. Three MFIA measurement sessions totaling 20 oranges with systematic variation of electrode configuration and freeze duration.
2. Identification of 180° as the optimal electrode configuration for core freeze detection (6.59× ratio, 10,607 Ω gap).
3. Mapping of the impedance–freeze duration relationship with 30-minute resolution, identifying the critical core-freeze transition at 2.5–3 hours.
4. Development of a portable AD5933 + Arduino prototype with custom firmware, validated against the MFIA reference.
5. SolidWorks enclosure design with integrated spring-loaded caliper for simultaneous electrode contact and diameter measurement.
6. Complete MATLAB analysis pipeline producing Bode, Nyquist, Cole-Cole parameter tracking, 3D surface, and statistical analysis plots.

6.2 Cole-Cole Fitting and Remaining CNLS Work

In an earlier draft the Cole-Cole parameters were read off by hand. For this version we used the course `colez` circle-fit routine instead (Sections 3.4.2, 4.2.4 and 4.2.5), which gives proper values for R_0, R_∞, α and f_c together with a fit error. For most of our oranges that error is only a few percent.

The circle fit has one weak spot, which our own data made obvious. It assumes the Nyquist plot is a single clean arc. Our 2-electrode measurements have an electrode-polarisation tail at low frequency, and for the most frozen oranges that tail takes over

completely, so the fit either reports a high error or cannot close the arc at all (the 10 h orange). A full complex non-linear least squares fit (CNLS, for example with `lsqcurvefit` or `fminsearch`) over the whole spectrum, with an extra circuit element for the electrode polarisation, would handle those cases better and would use the phase information as well as the magnitude. Figure 44 shows how a clean CNLS fit looks on simulated data. Moving to a 4-electrode setup (Section 6.5) would also remove the polarisation tail at the source.

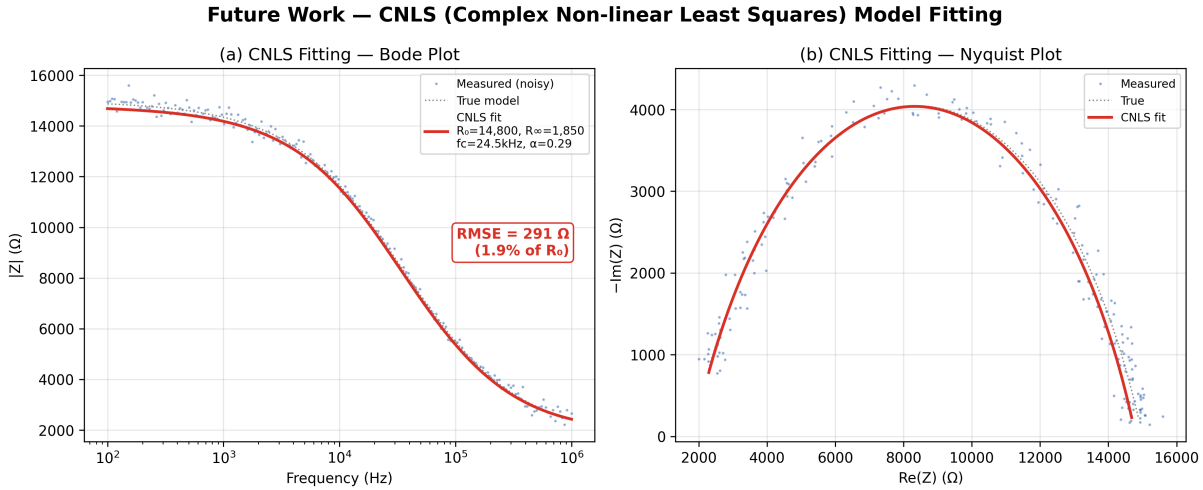


Figure 44: Simulated CNLS fitting. (a) Bode plot: noisy simulated data (dots), the true model (dashed), and the CNLS fit (red); the fit recovers the parameters to within about 1.5% despite the noise. (b) Same in Nyquist form. A full CNLS fit would extract R_0, R_∞, α and f_c automatically for every orange.

6.3 Machine Learning Classification

With sufficient data (e.g., 50+ oranges per class), a machine learning classifier could be trained using multiple impedance features. Figure 45 shows a simulation of the expected performance.

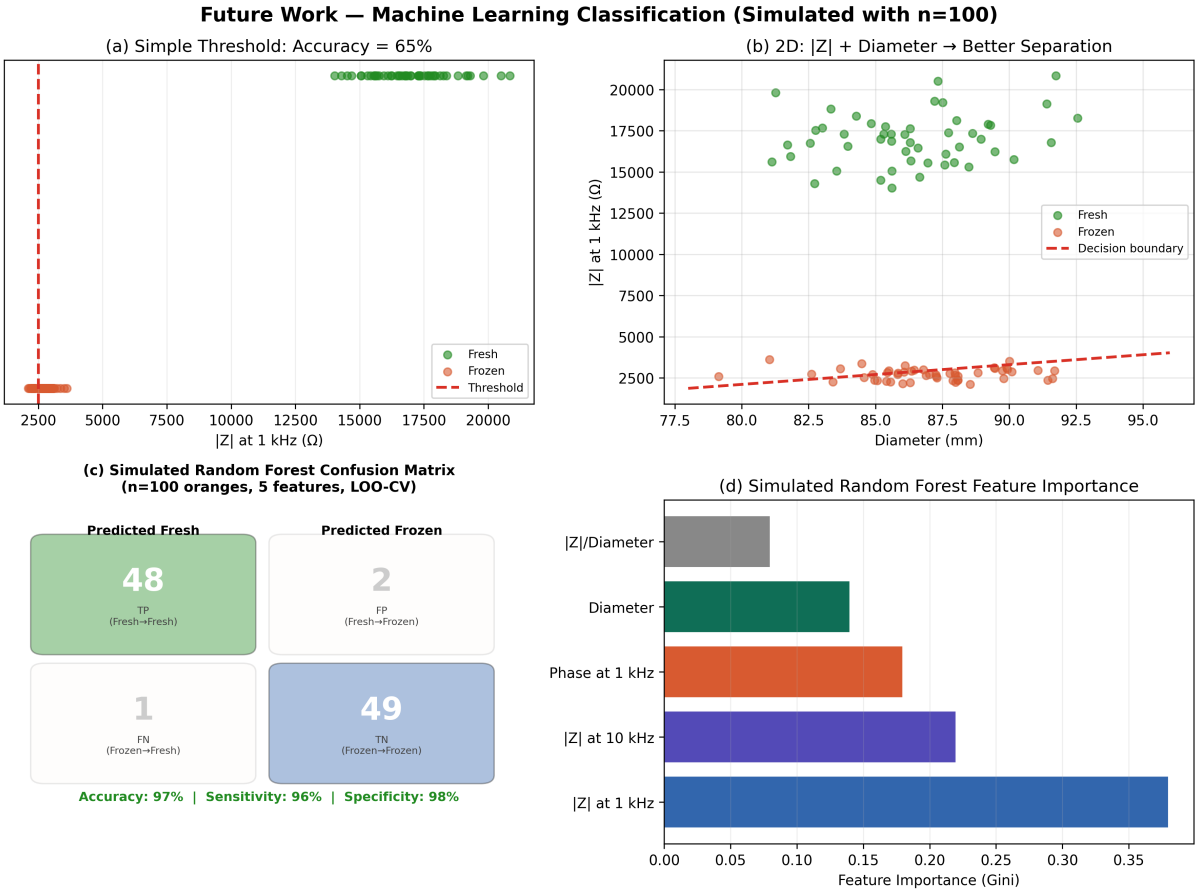


Figure 45: Simulated ML classification with n=100 oranges. (a) Simple threshold: some overlap causes misclassification. (b) 2D features (—Z— + diameter): linear decision boundary provides better separation. (c) Random Forest confusion matrix (simulated LOO-CV): 97% accuracy with 5 features. (d) Feature importance ranking: —Z— at 1 kHz is most important, but diameter and phase add discriminating power.

Proposed ML pipeline for future work:

- **Features:** $|Z|$ at 1 kHz, $|Z|$ at 10 kHz, phase at 1 kHz, diameter, $|Z|$/diameter
- **Extended features (from CNLS):** R_0, R_∞, $(R_0 - R_\infty)$, f_c, α
- **Algorithm:** Random Forest (robust to small datasets, interpretable feature importance)
- **Validation:** Leave-one-out cross-validation (appropriate for small n)
- **Data requirement:** Minimum 30 oranges per class (fresh, frozen <2h, frozen 2–4h, frozen >4h)

6.4 Physics-Informed Neural Networks

For applications with very limited training data (as in our case), Physics-Informed Neural Networks (PINNs) offer a promising approach, as discussed in the course lecture (slides 31–35). Figure 46 shows the proposed architecture.

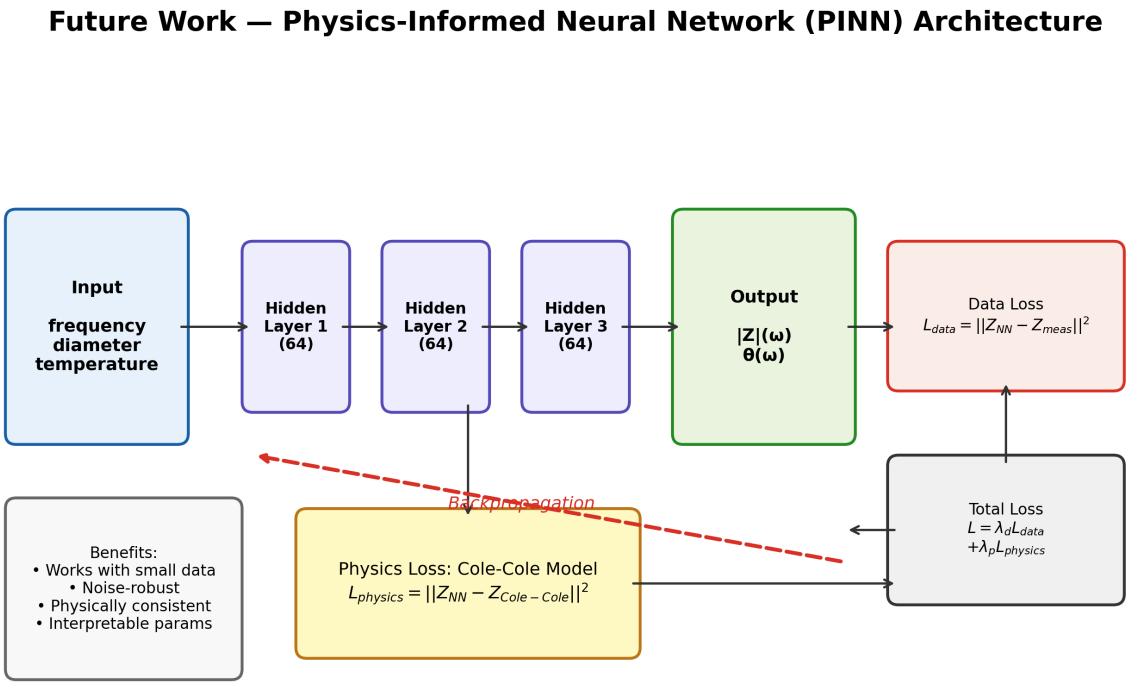


Figure 46: Proposed PINN architecture for impedance-based classification. The network takes frequency, diameter, and temperature as inputs and predicts $|Z|(\omega)$ and $\theta(\omega)$. The loss function combines data fidelity (L_{data}) with physics consistency ($L_{physics}$, enforcing the Cole-Cole model). This regularization reduces the data requirement and ensures physically plausible predictions, particularly important in the small-data regime of our application.

The PINN approach is particularly relevant because: (1) our dataset is small (20 oranges), making pure data-driven ML prone to overfitting, (2) we have a well-established physical model (Cole-Cole), and (3) the physics constraint acts as a regularizer that improves generalization. Recent work by Crandall et al. (2025) demonstrated this approach for blood-pressure estimation from wrist EIS using Navier-Stokes and Maxwell-Fricke equations [5].

6.5 Additional Future Improvements

- **AD5933 artifact mitigation:** Start the sweep at a dummy frequency (e.g., 500 Hz) and discard the first data point, allowing 1 kHz measurements from the portable system.
- **4-electrode configuration:** Eliminate electrode contact impedance contribution for more accurate tissue-only measurements.
- **Multi-variety testing:** Validate across different orange varieties (Navel, Valencia, Blood) and other citrus (lemons, grapefruits).
- **Repeatability study:** 3+ measurements per orange with electrode re-insertion to quantify intra-orange variability.
- **3D-printed enclosure:** Fabricate and test the SolidWorks design with actual hardware integration.

- **Industrial integration:** Adapt the caliper mechanism for conveyor-belt mounting in packing houses.
- **Open-Short-Load calibration:** Implement the three-reference Bao method (course lecture slide 17–19) for the AD5933 to improve accuracy beyond the single-reference gain factor.

7 Conclusions

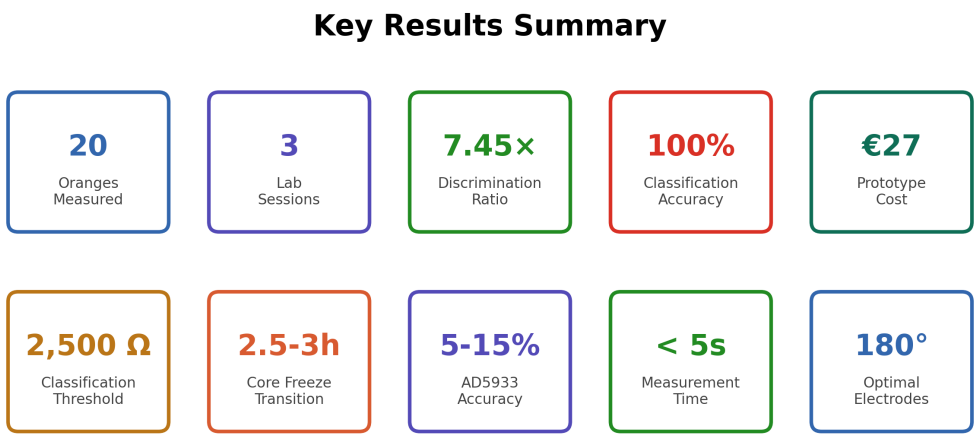


Figure 47: Summary of key results from the frozen orange detection project.

This project demonstrated that Electrical Impedance Spectroscopy is an effective, non-destructive, real-time method for detecting previously frozen oranges. The principal conclusions are:

1. **Clear separation:** Fresh and frozen oranges are unambiguously distinguished by their impedance at 1 kHz, with a discrimination ratio of 7.45× (vertical) / 6.59× (180°) and zero overlap between populations (n=20).
2. **Physical mechanism confirmed:** Freezing ruptures the cell membranes, which act as distributed capacitors. The measured $|Z|$ at 1 kHz drops from about 15,000 Ω (fresh) to about 2,300 Ω (frozen 24h+), and the Cole-Cole arc diameter from the circle fit collapses from about 35,900 Ω to under 1,000 Ω, which we can read straight off the Nyquist plot.
3. **Optimal measurement configuration:** 180° electrode placement at the equator with 2 cm insertion depth provides the best detection of core freeze damage, with a 10,607 Ω gap between fresh and frozen populations at 1 kHz.

4. **Core freeze transition timing:** The critical impedance drop occurs between 2.5 and 3 hours of freezing at −18°C for oranges of ∼87 mm diameter, significantly faster than the 7 hours predicted by the classical Plank equation, attributed to forced-air circulation in domestic freezers.

5. **Simple and robust classification:** A threshold of 2,500 Ω at 1 kHz achieves 100% accuracy (sensitivity = specificity = 100%) on the combined 20-orange dataset across three measurement sessions.

6. **Diameter as confounding variable:** Orange-to-orange size variation is the primary source of impedance variability among non-frozen samples. Diameter normalization ($|Z|/D$) partially mitigates this, and automatic diameter measurement has been integrated into the enclosure design.

7. **Portable system feasibility:** The AD5933 + Arduino Nano prototype provides impedance measurements at ∼€27 total cost (555× cheaper than the MFIA reference), with accuracy sufficient for binary classification despite a known 1 kHz startup artifact resolved by using 10 kHz.

8. **Phase provides additional discrimination:** The impedance phase transitions from −28° (capacitive, intact membranes) to −10° (resistive, ruptured membranes), providing a second independent classification feature that could improve ML-based classification in future work.

The combination of high discrimination ratio, zero classification errors, non-destructive measurement, fast response (<5 s), and low cost (€27) makes EIS a compelling technology for frozen orange detection in industrial, professional, and research settings.

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